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Mehta et al.

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(54) SYSTEMS, DEVICES, AND METHODS FOR THE RETRIEVAL OF AN IMPLANT IN THE PROSTATIC URETHRA

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(73) Assignee: **ZENFLOW, INC.**, San Francisco, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 952 days.

(21) Appl. No.: **17/495,996**

(22) Filed: **Oct. 7, 2021**

(65) **Prior Publication Data**

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Related U.S. Application Data

(60) Provisional application No. 63/089,205, filed on Oct. 8, 2020.

(51) **Int. Cl.**

A61F 2/95	(2013.01)
A61B 17/00	(2006.01)
A61F 2/04	(2013.01)

(52) **U.S. Cl.**

CPC ***A61F 2/04*** (2013.01); ***A61B 17/00234*** (2013.01); *A61B 2017/00274* (2013.01); *A61F 2002/047* (2013.01)

(58) **Field of Classification Search**

CPC A61F 2002/9528; A61F 2002/9534; A61F 2002/047; A61F 2002/046–2002/048; A61F 2/95; A61F 2002/9505–2002/9511

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,941,895 A *	8/1999	Myler	A61F 2/95	606/198
11,571,290 B2 *	2/2023	Bachar	A61F 2/04	
12,076,226 B2 *	9/2024	Bachar	A61F 2/9517	
2002/0120277 A1 *	8/2002	Hauschild	A61F 2/95	606/108
2002/0161427 A1 *	10/2002	Rabkin	A61F 2/95	623/1.11

(Continued)

FOREIGN PATENT DOCUMENTS

CN	102 438 533 B	11/2014
CN	108836587 A	11/2018

(Continued)

OTHER PUBLICATIONS

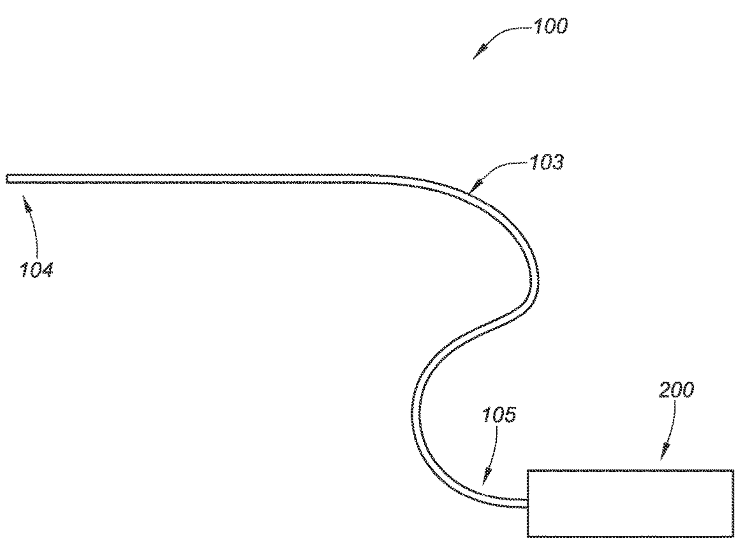
EP, 21878507.9 Extended Search Report, Oct. 10, 2024.
WO, PCT/US21/53904 ISR and Written Opinion, Feb. 17, 2022.
JP, 2023-521338 Office Action, Jul. 1, 2025.

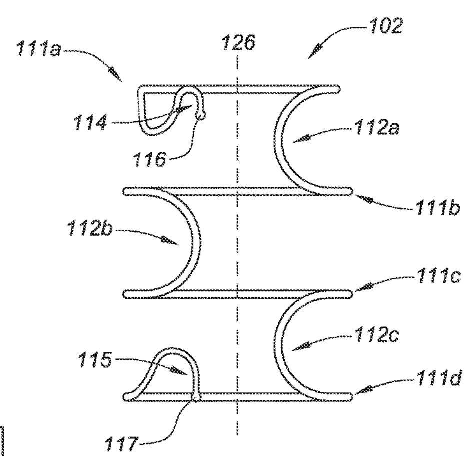
Primary Examiner — Alvin J Stewart
(74) *Attorney, Agent, or Firm* — ONE LLP

(57) **ABSTRACT**

Systems, devices, and methods are provided for retrieval of an implant from the prostatic urethra. Embodiments of retrieval systems can include a device for insertion into the patient and a proximal control device for use in grasping a portion of the implant and withdrawing the implant into a lumen of the retrieval system.

12 Claims, 37 Drawing Sheets





(56) **References Cited**

U.S. PATENT DOCUMENTS

2002/0188344	A1*	12/2002	Bolea	A61F 2/95 623/1.11
2005/0131423	A1*	6/2005	Yachia	A61F 2/95 606/108
2005/0131515	A1*	6/2005	Cully	A61F 2/07 623/1.13
2007/0186933	A1	8/2007	Domingo et al.	
2010/0094327	A1*	4/2010	Milsom	A61B 17/0218 606/191
2011/0054519	A1	3/2011	Neuss	
2015/0257908	A1	9/2015	Chao et al.	
2016/0030047	A1	2/2016	Allen et al.	
2016/0206449	A1*	7/2016	Mort	A61F 2/86
2016/0346107	A1*	12/2016	Matthison-Hansen	A61B 17/30
2017/0065299	A1	3/2017	Gillespie et al.	
2019/0038443	A1	2/2019	Sicotte et al.	
2019/0099280	A1	4/2019	Ben Muvhar	
2019/0117423	A1	4/2019	Chao et al.	
2019/0175199	A1	6/2019	Girdhar et al.	
2019/0307548	A1	10/2019	Sicotte et al.	
2020/0038213	A1	2/2020	Bly et al.	
2020/0164188	A1	5/2020	Chang et al.	
2020/0323618	A1	10/2020	Bly et al.	
2021/0137711	A1*	5/2021	Fukumoto	A61F 2/9522
2021/0145619	A1	5/2021	Bly et al.	
2021/0161641	A1*	6/2021	Bachar	A61F 2/962
2021/0161642	A1*	6/2021	Jen	A61B 1/0008
2021/0251789	A1*	8/2021	Ouyang	A61B 1/00052
2021/0275335	A1	9/2021	Sicotte et al.	
2022/0110737	A1*	4/2022	Mehta	A61F 2/95
2024/0115405	A1*	4/2024	Kadlec	A61F 2/86

FOREIGN PATENT DOCUMENTS

EP	1 917 922	A1	5/2008	
JP	H1076015	A *	3/1998	A61B 17/221
WO	WO 2017/184887	A1	10/2017	
WO	WO 2019/222481	A1	11/2019	
WO	WO 2020/142846	A1	7/2020	
WO	WO 2021/101951	A1	5/2021	

* cited by examiner

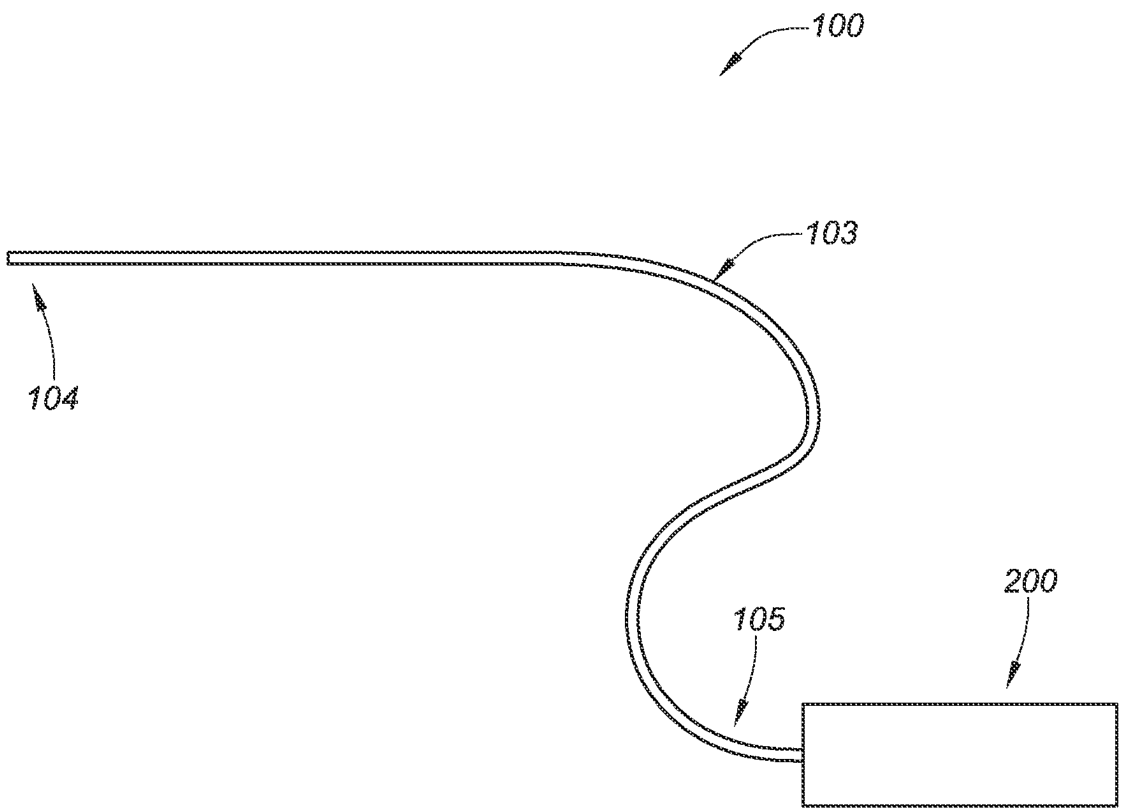


FIG. 1A

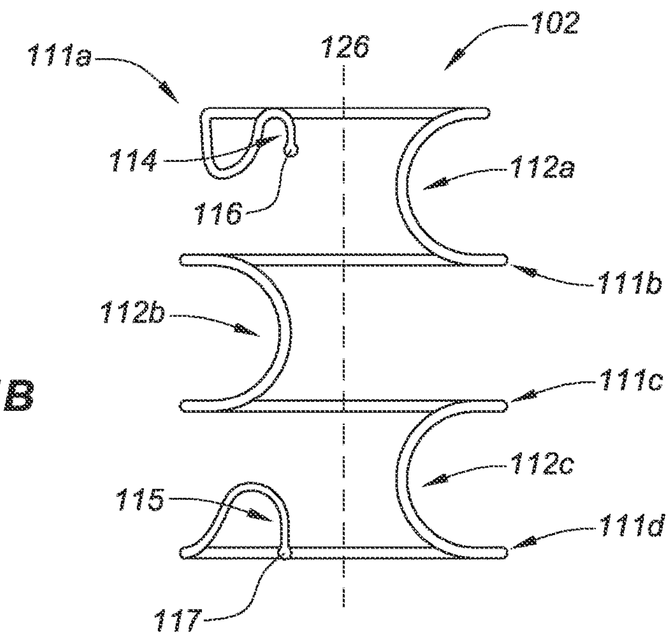


FIG. 1B

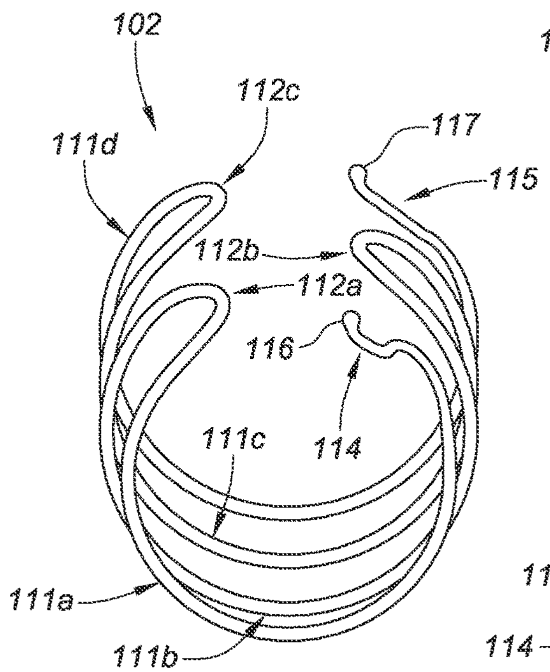


FIG. 1C

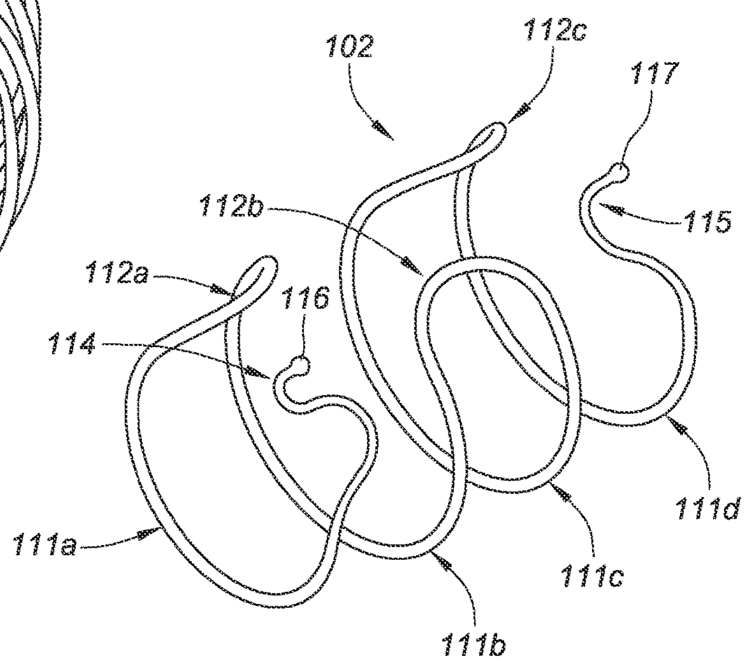


FIG. 1D

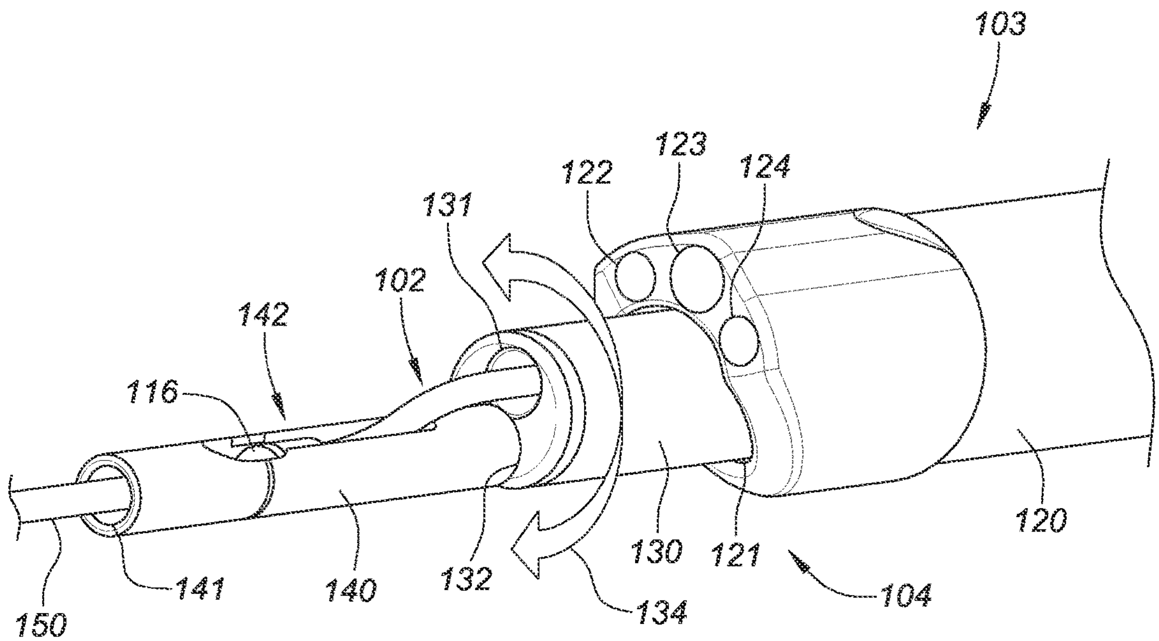

103
122
123
124
131
102
142
116
150
141
140
132
134
130
121
104
120

FIG. 2A

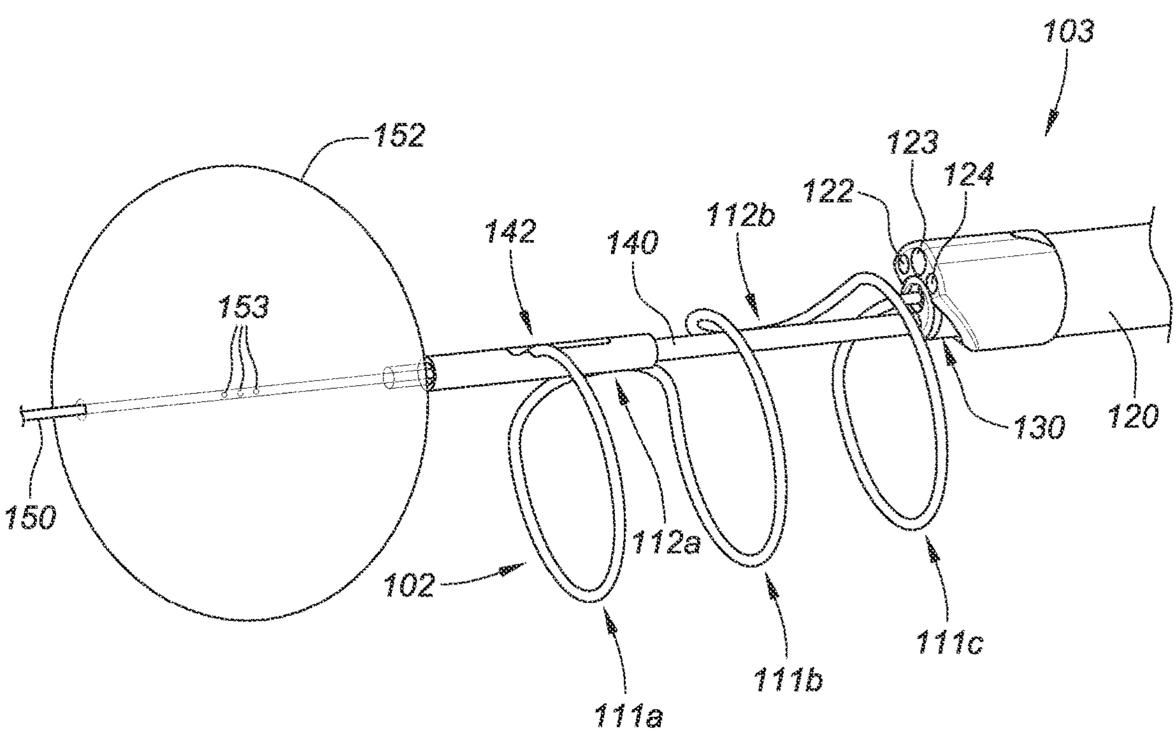
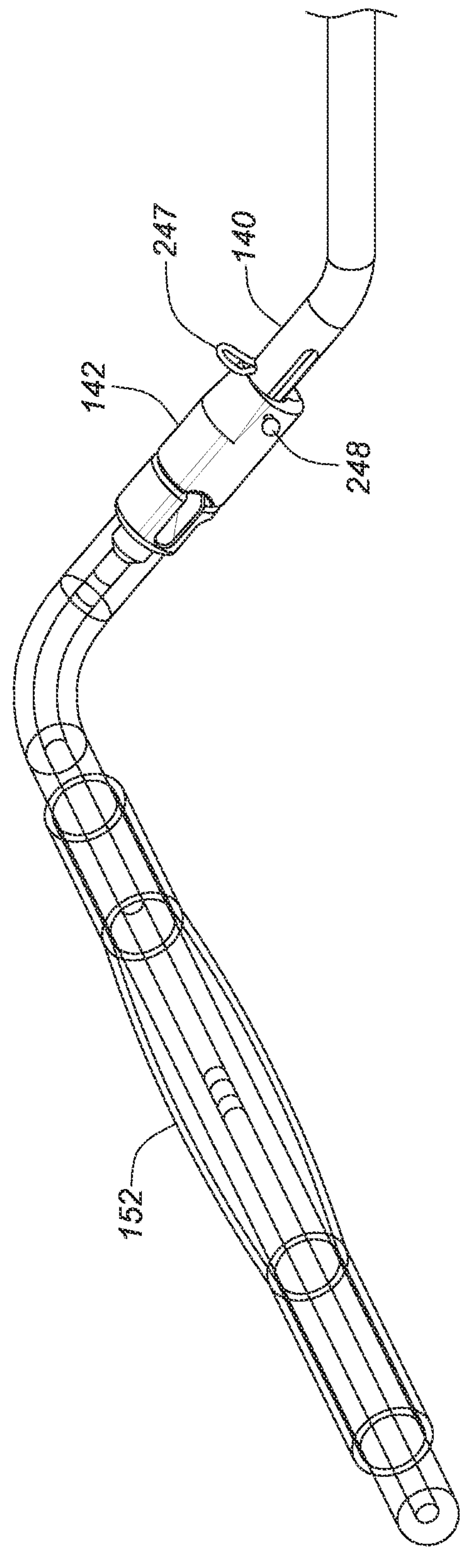

103
152
123
124
122
112b
142
140
153
150
112a
102
111a
111b
111c
130
120

FIG. 2B



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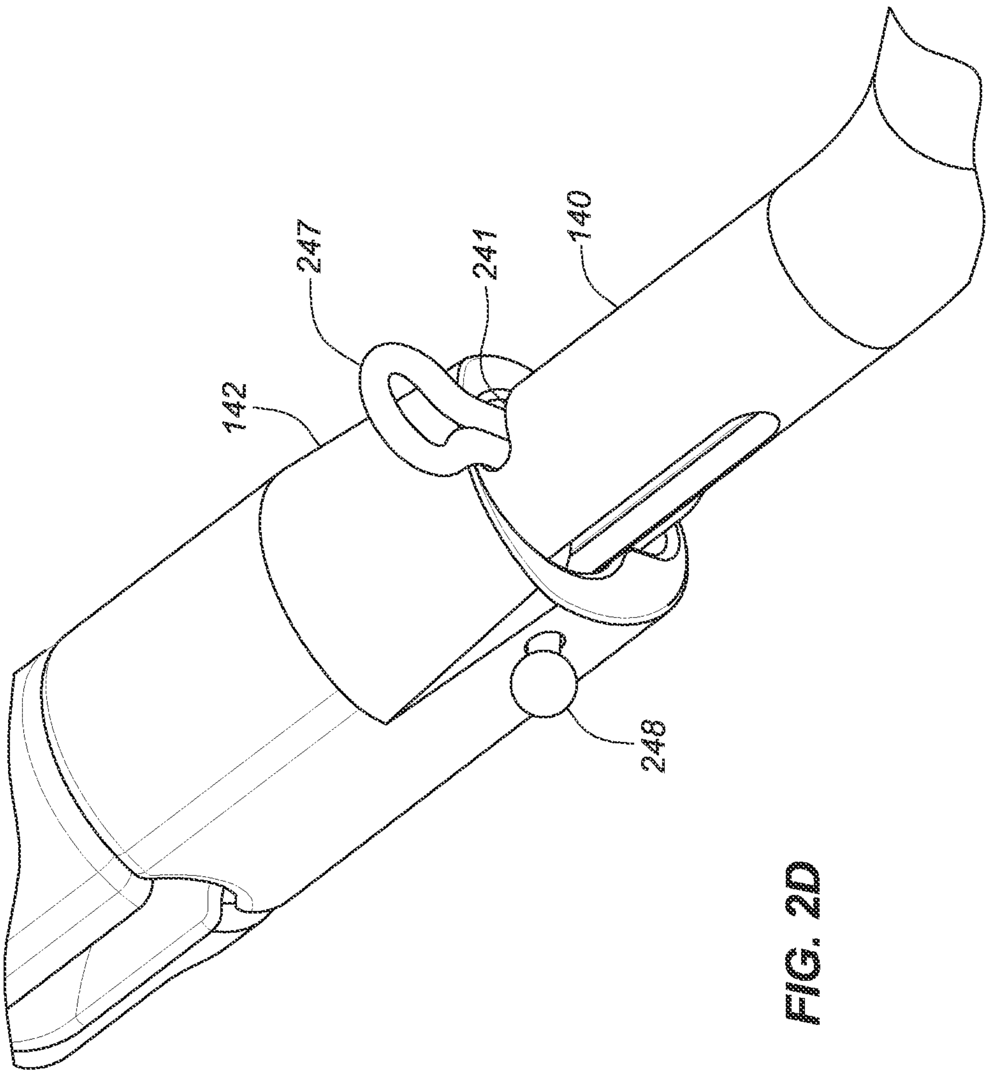


FIG. 2D

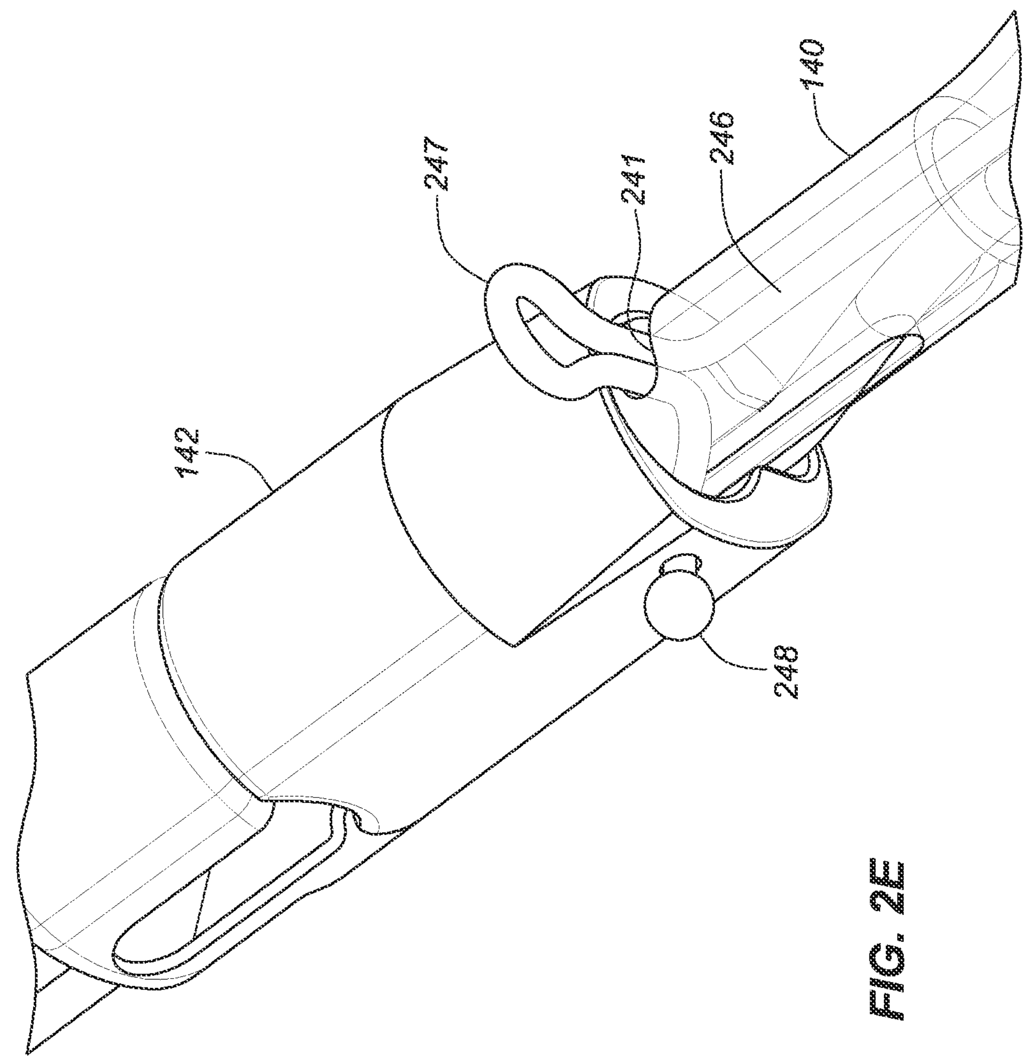


FIG. 2E

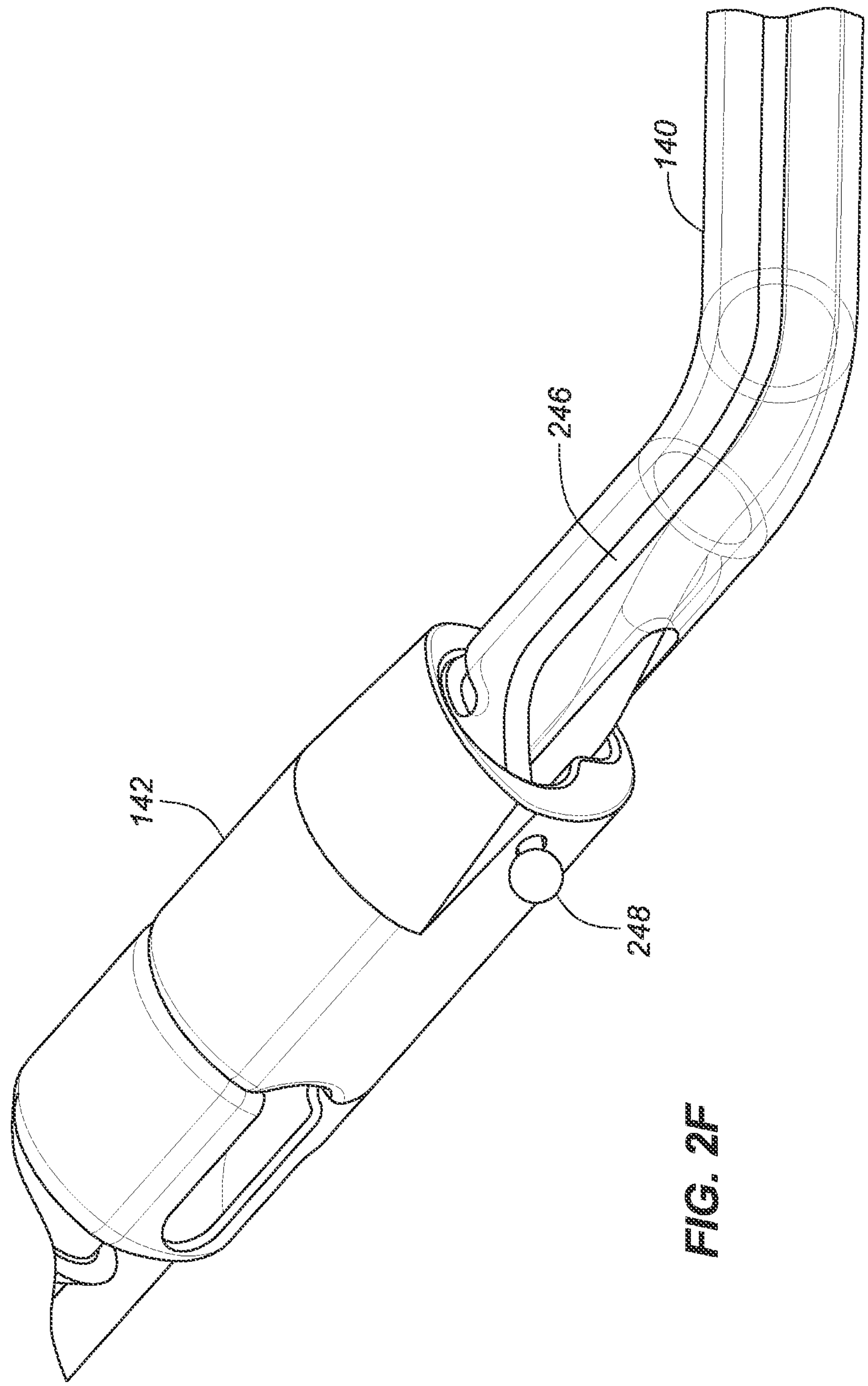


FIG. 2F

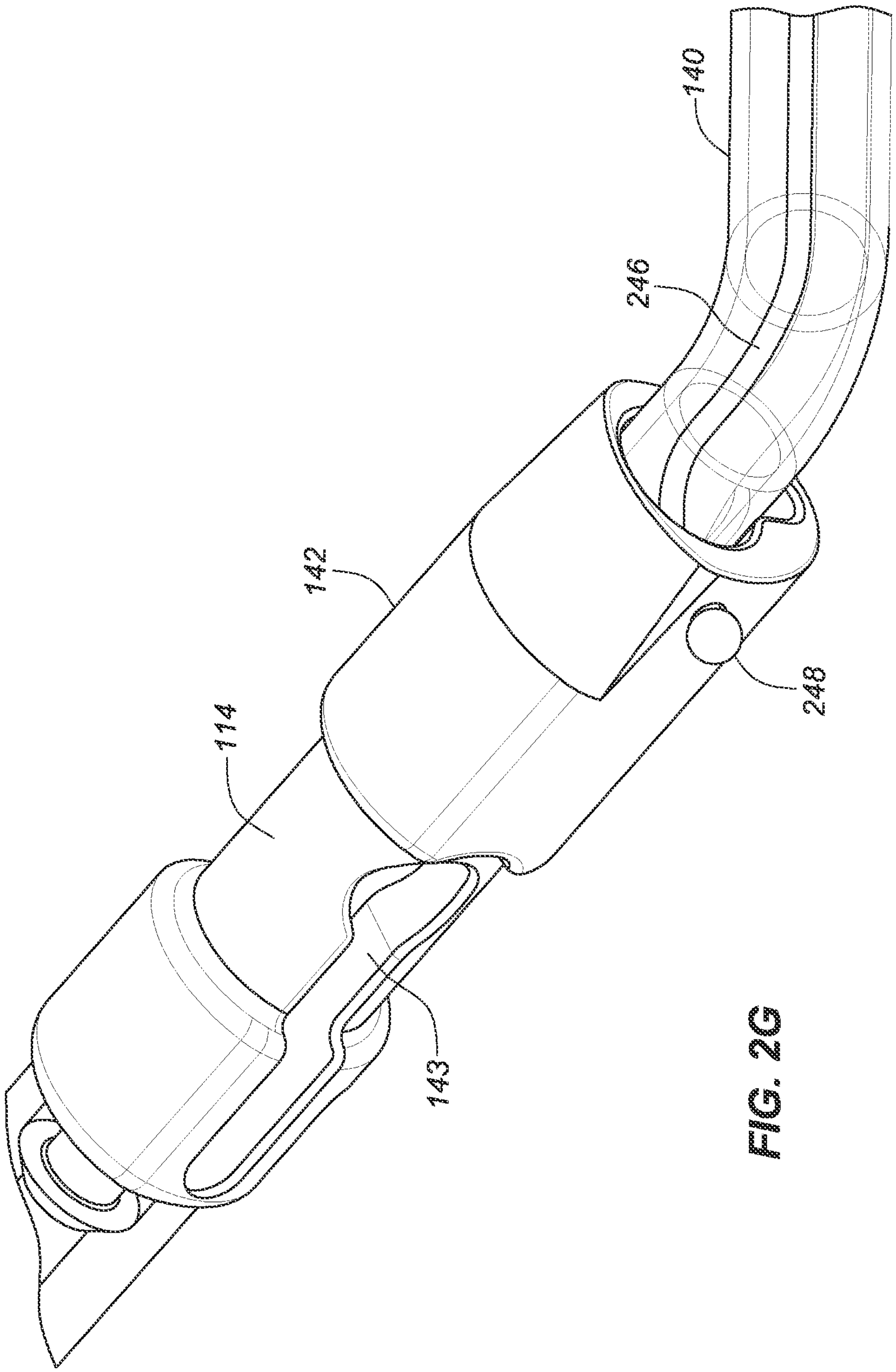


FIG. 2G

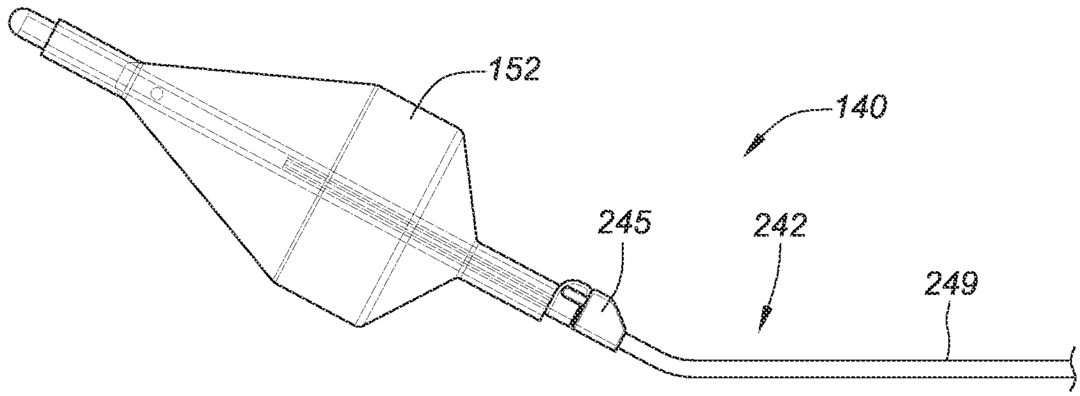


FIG. 2H

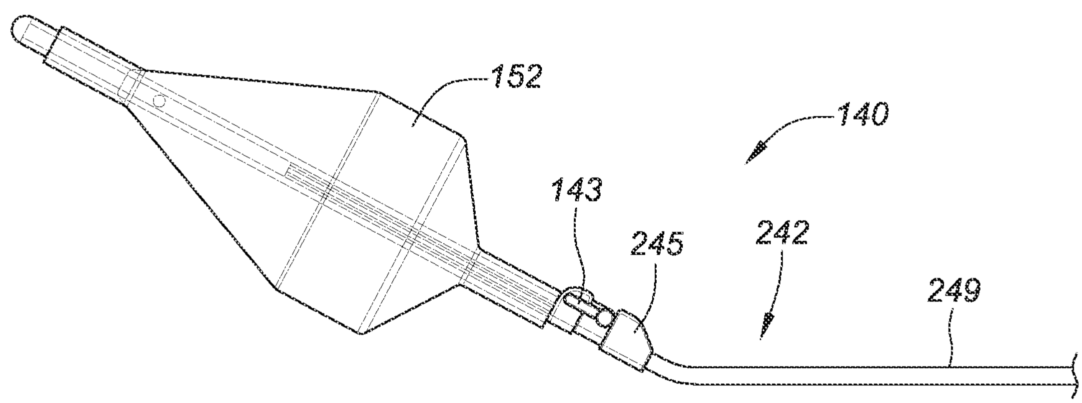


FIG. 2I

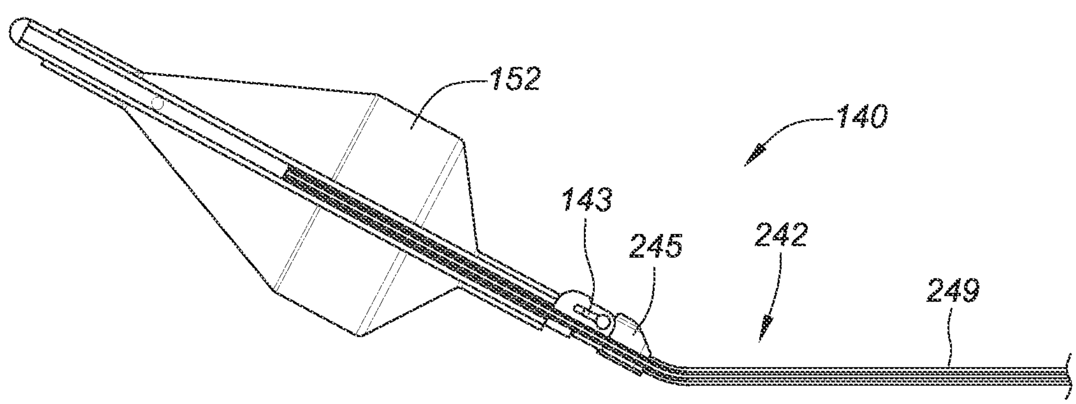


FIG. 2J

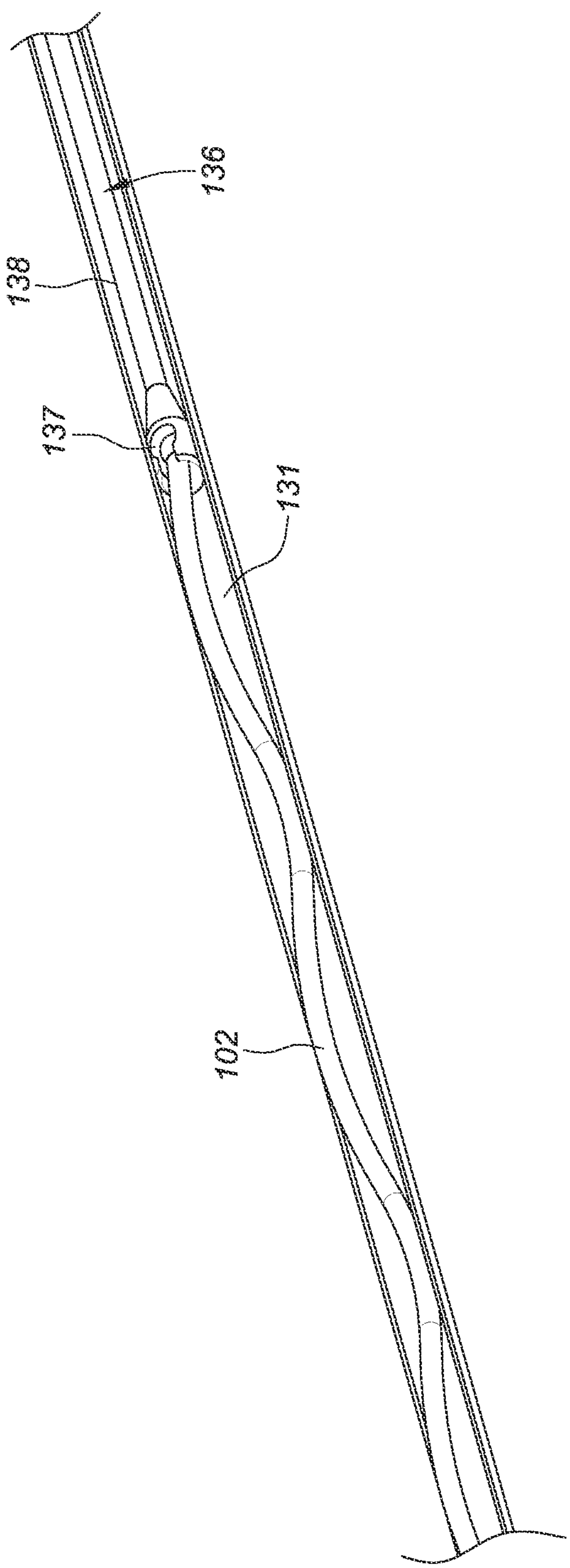


FIG. 3A

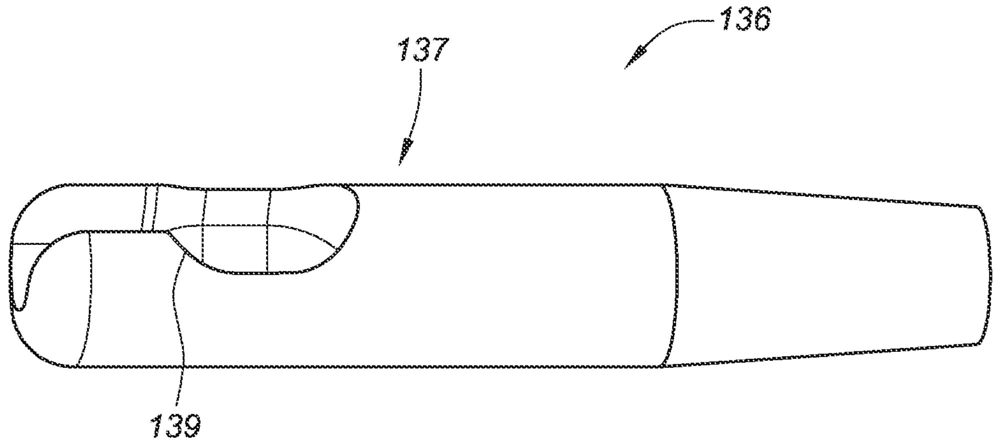


FIG. 3B

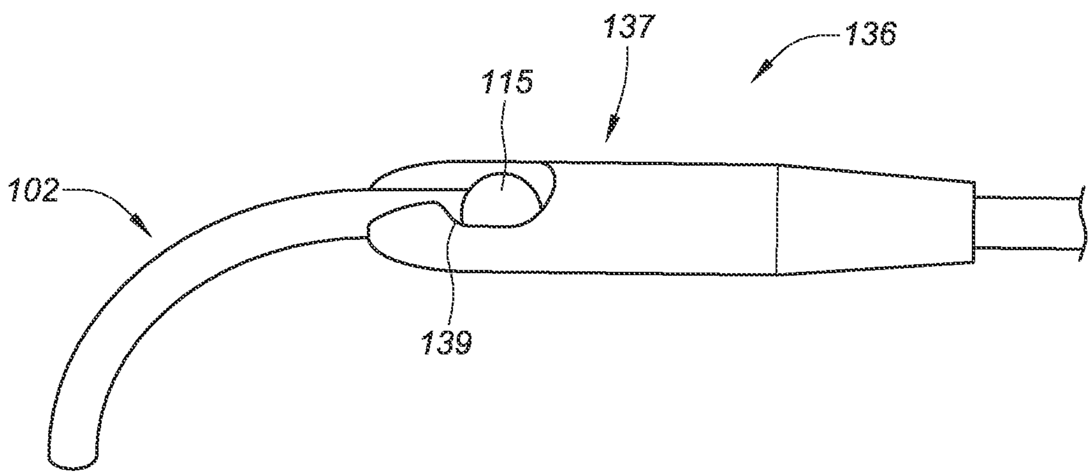


FIG. 3C

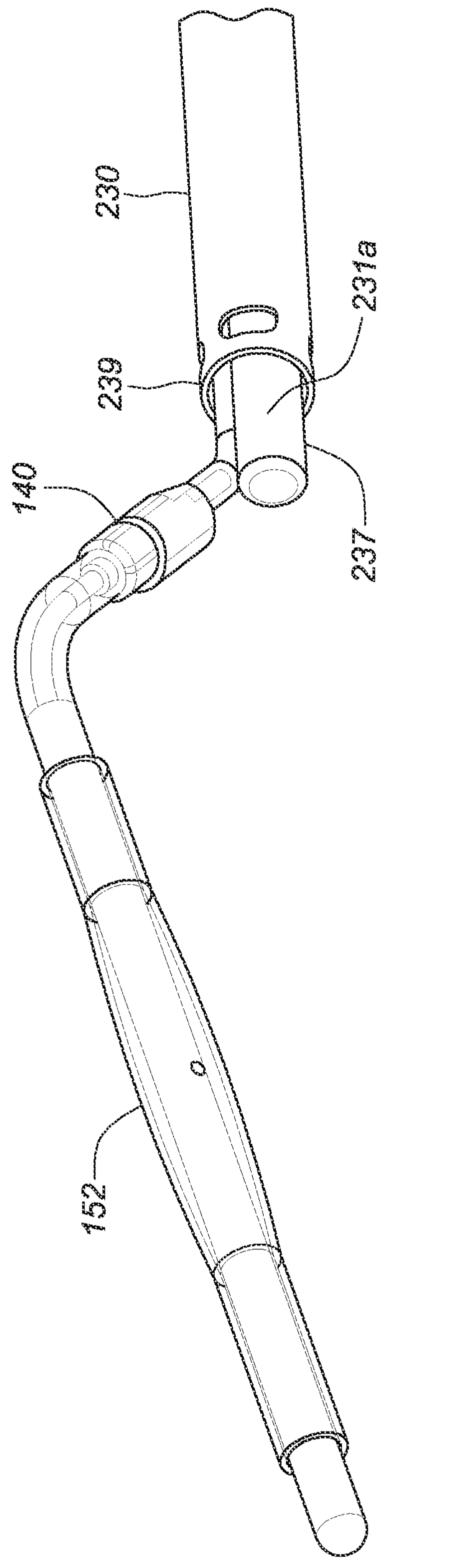


FIG. 4A

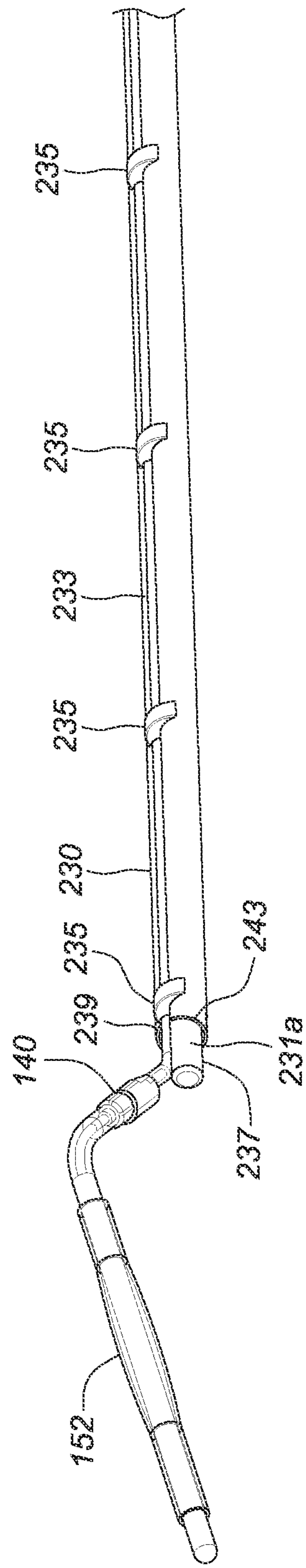
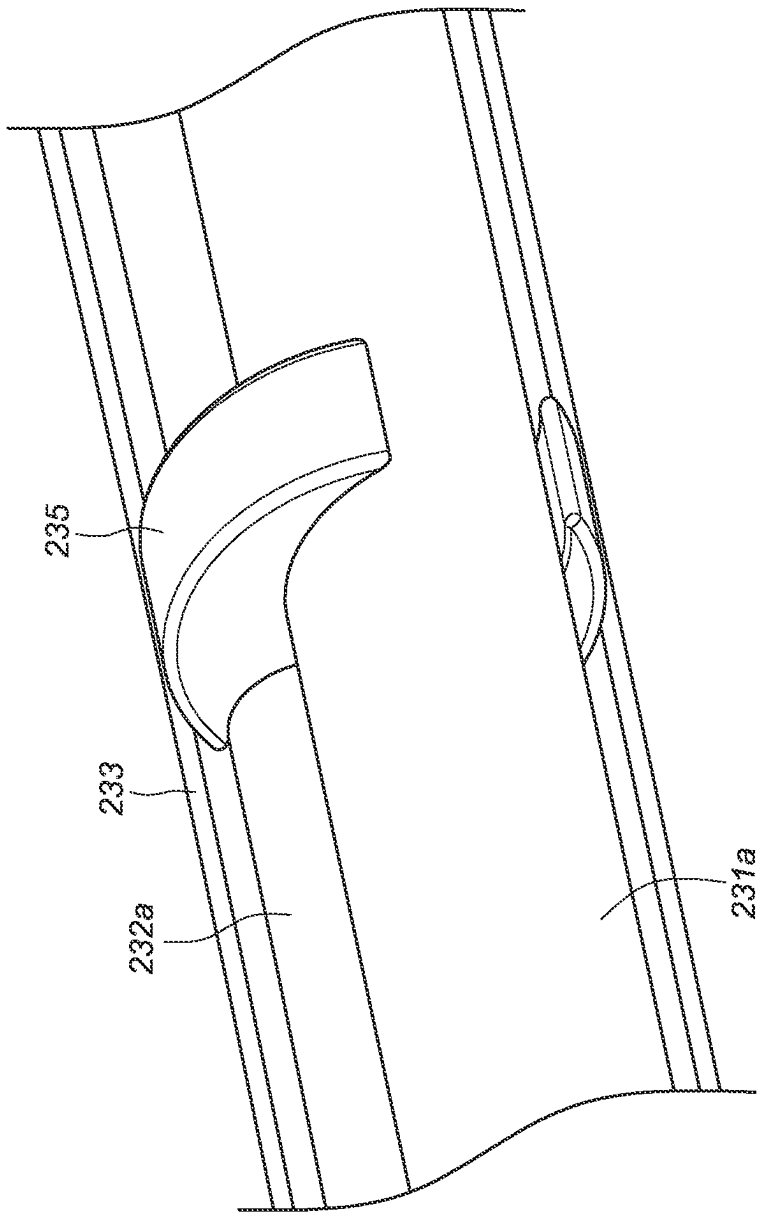


FIG. 4B



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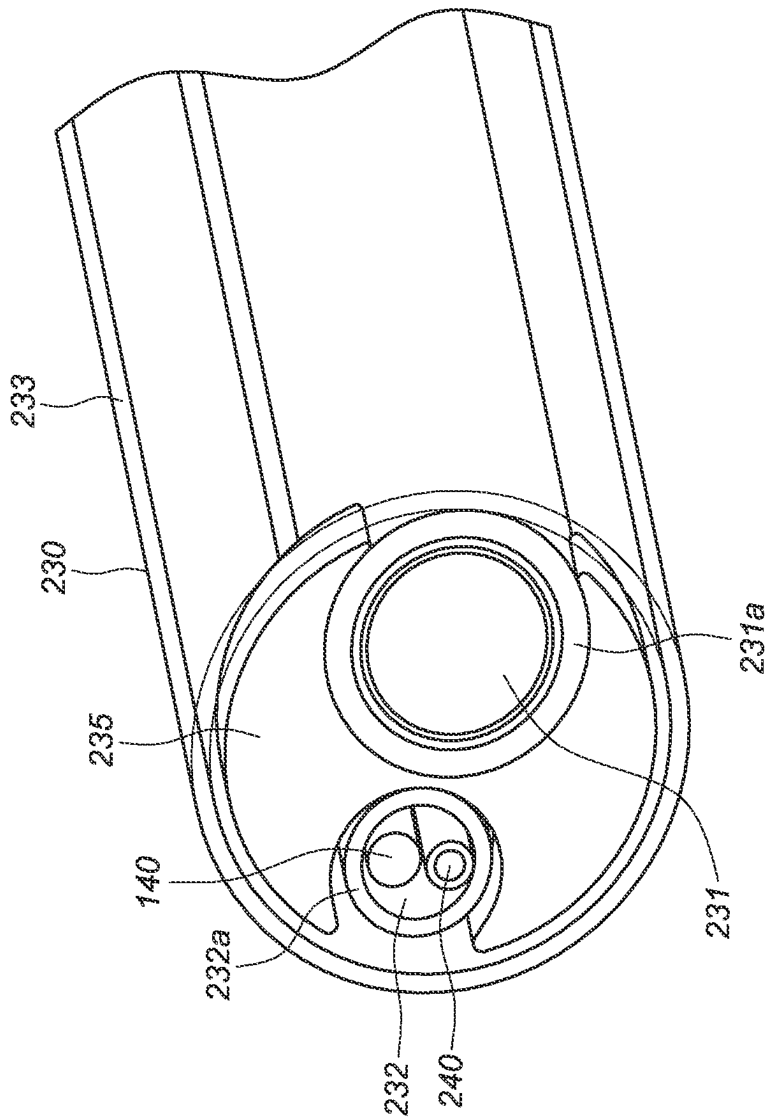


FIG. 4D

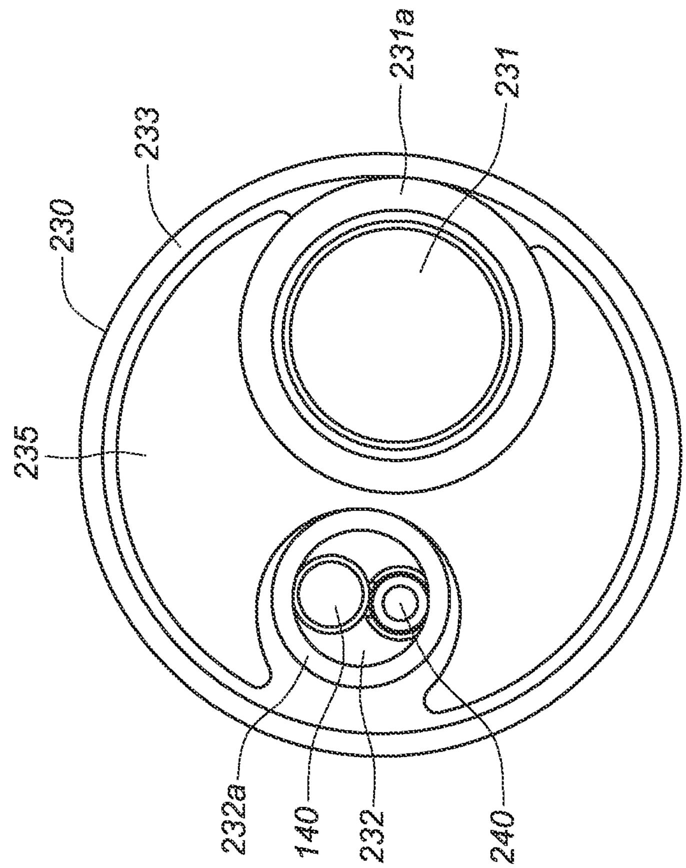


FIG. 4E

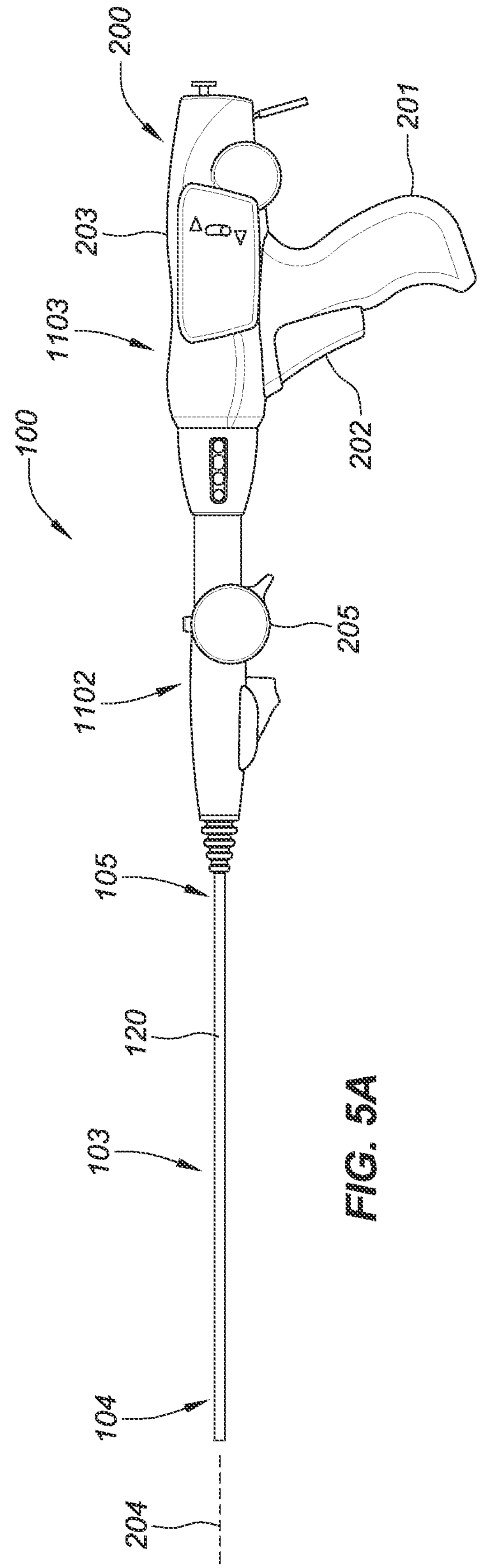

200
201
203
1103
202
100
205
1102
105
120
103
104
204

FIG. 5A

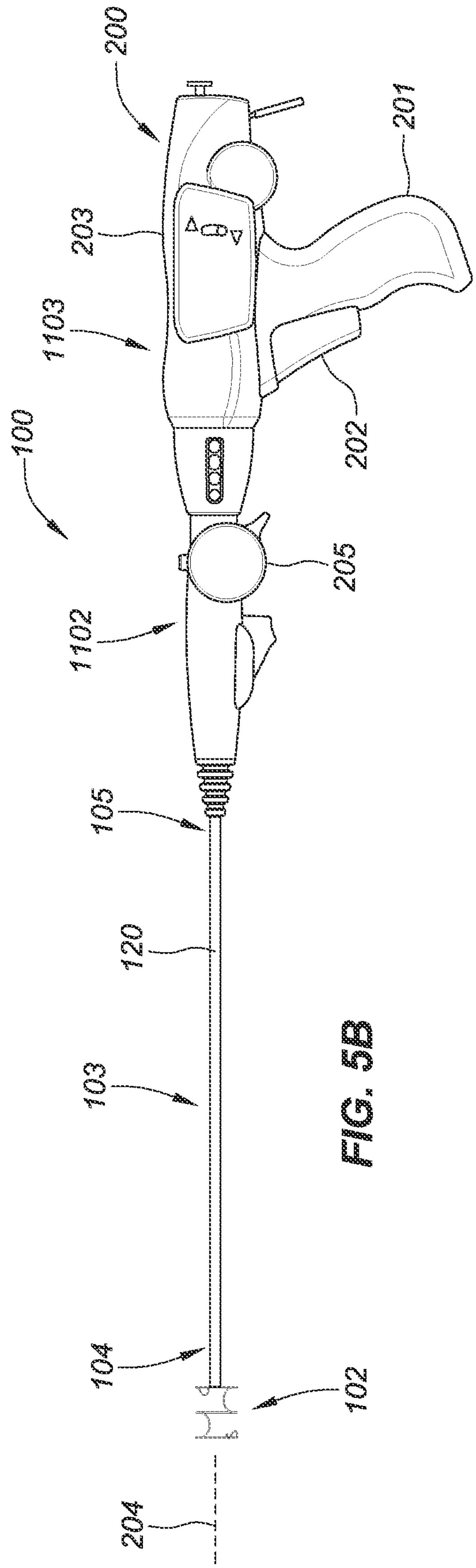

200
201
203
1103
202
100
205
1102
105
120
103
104
102
204

FIG. 5B

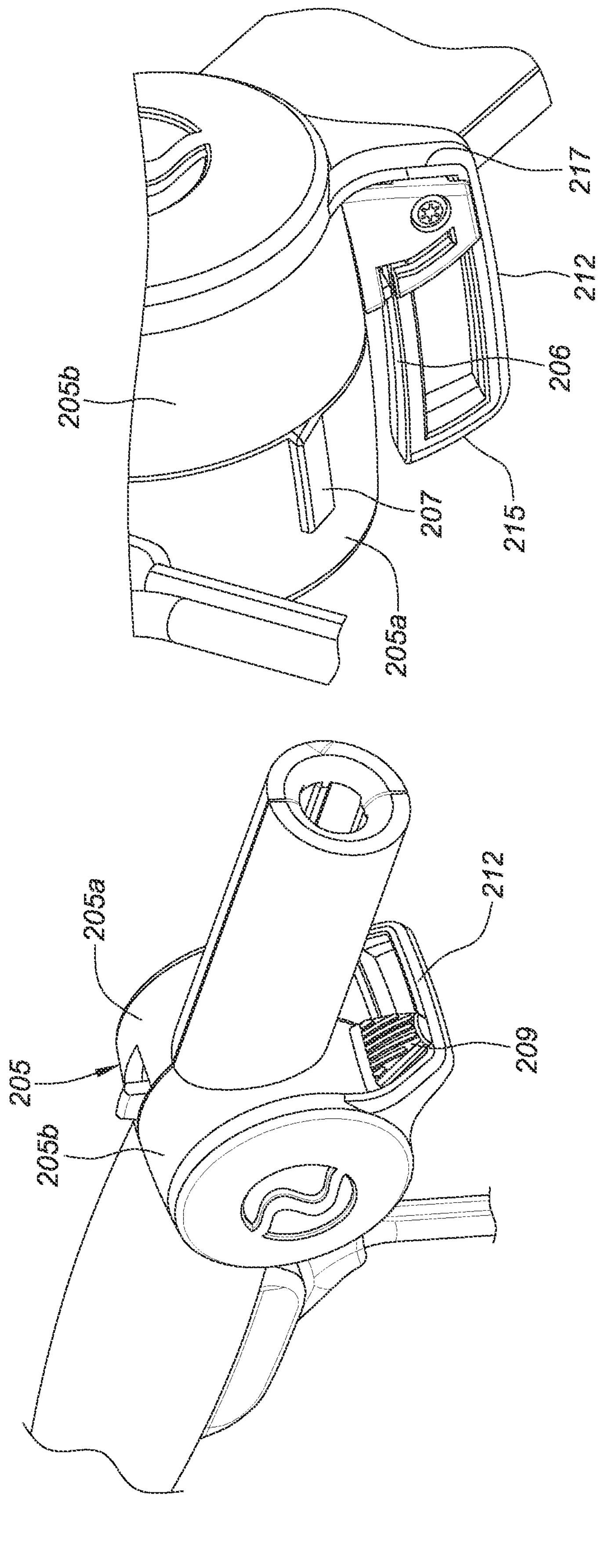


FIG. 5C

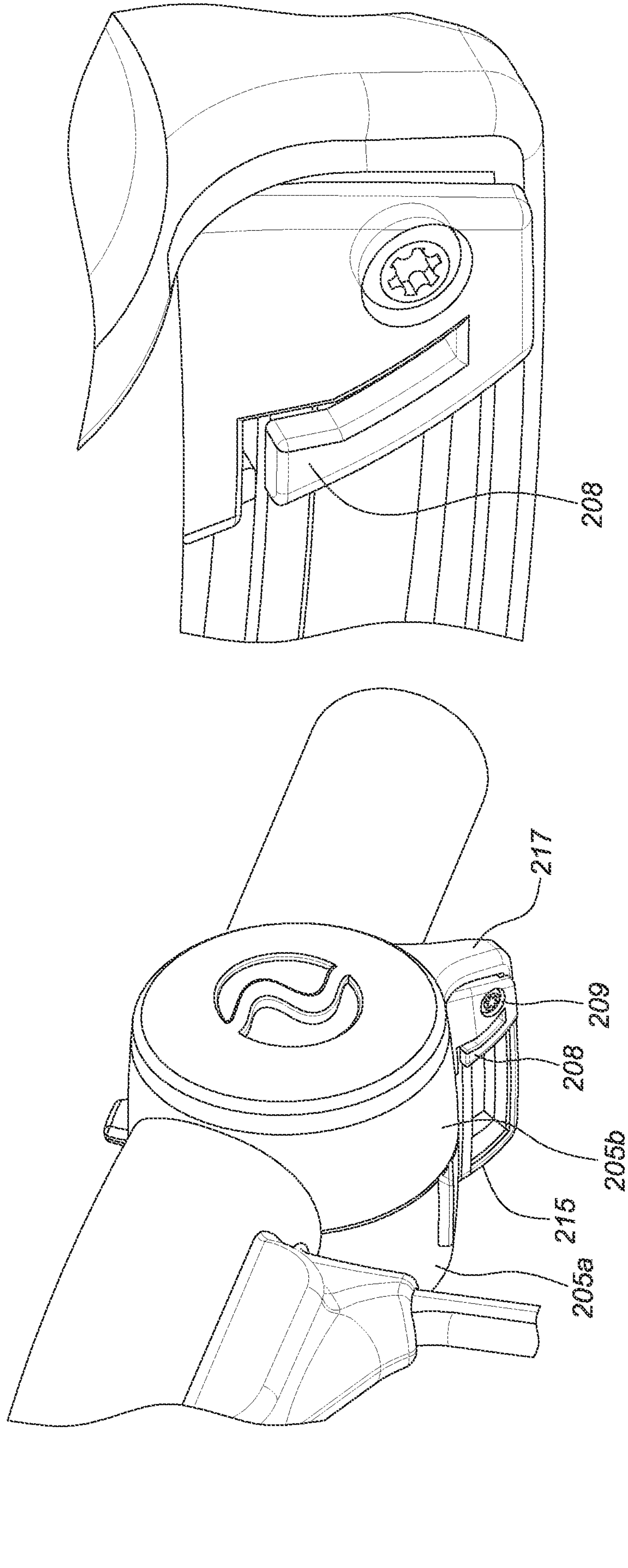


FIG. 5D

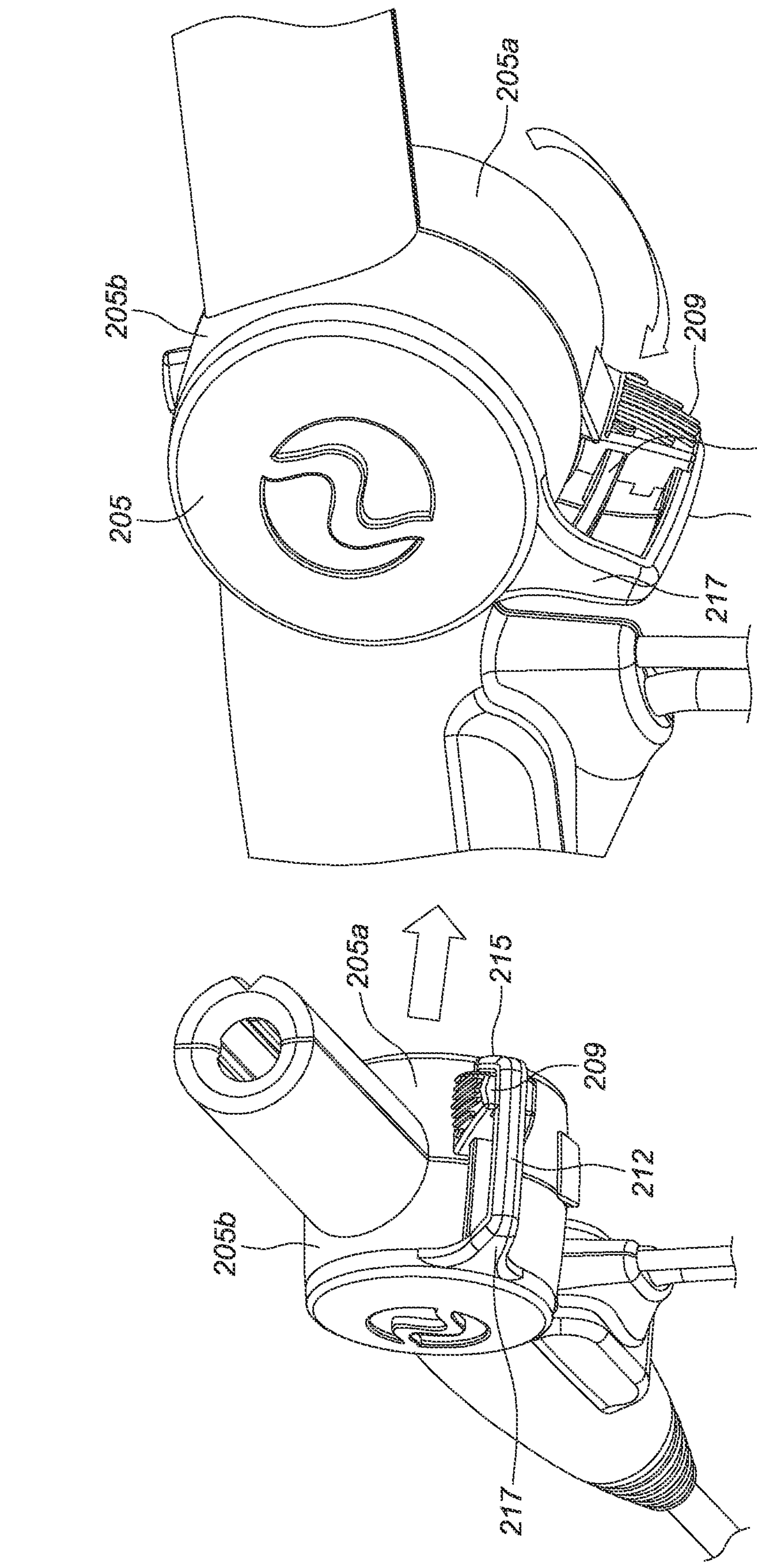


FIG. 5E

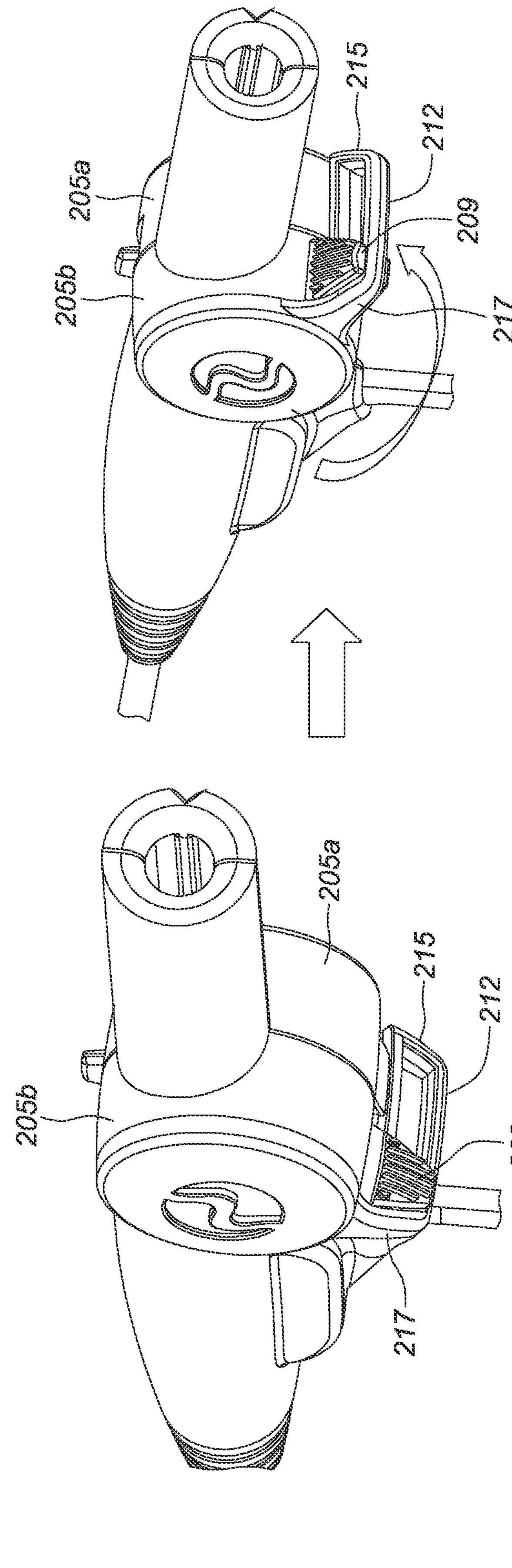


FIG. 5F

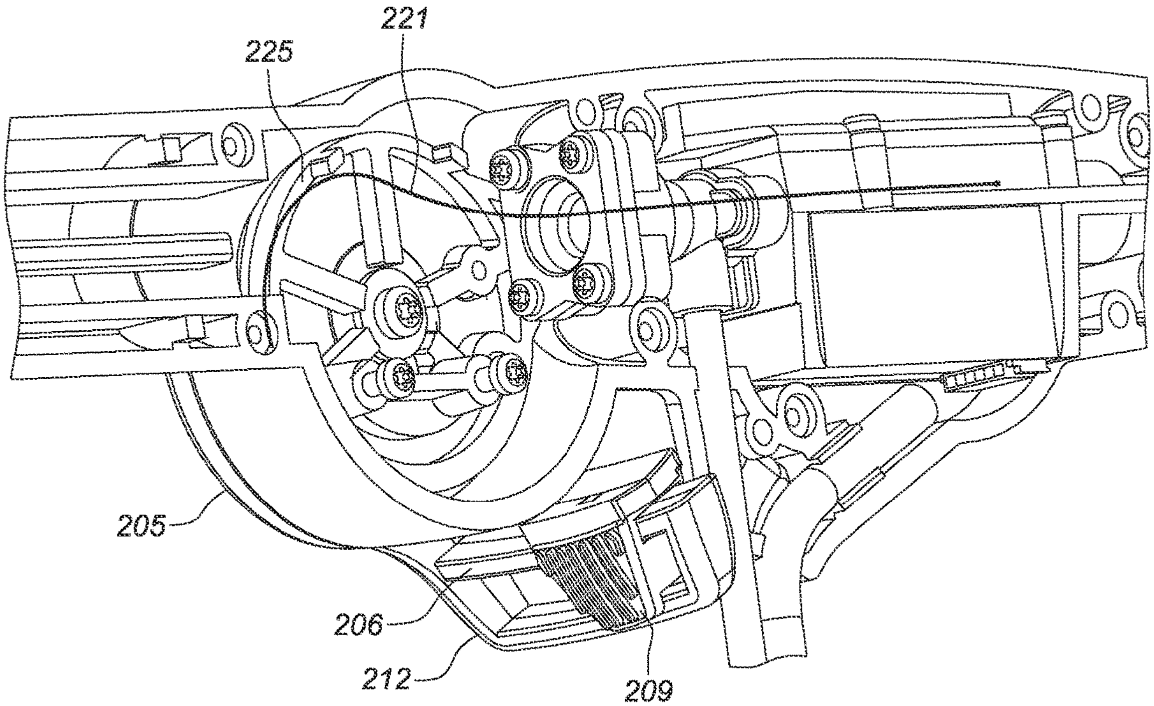


FIG. 5G

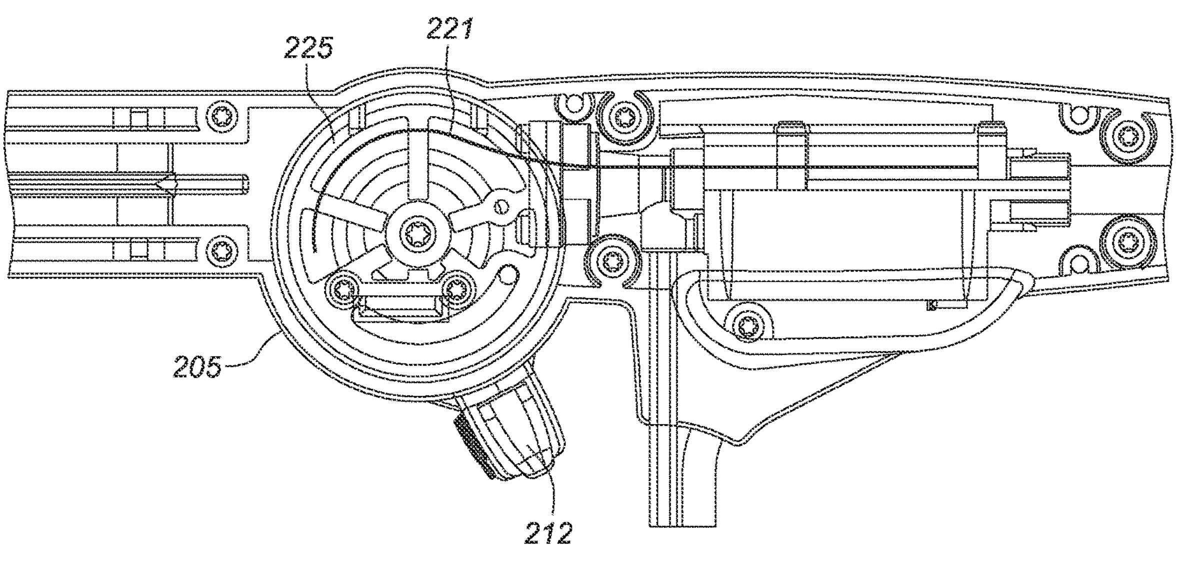


FIG. 5H

1000

Anchor delivery member 150 can be advanced distally with respect to the remainder of delivery device 103 and used to deploy anchor 152 within the bladder — 1002

↓

Anchor 152 can be held in tension against the bladder wall by exertion of a proximally directed force — 1004

↓

Distal control member 140 and inner shaft 130 can be distally advanced from within outer shaft 120 if they have not already — 1006

↓

A first ring-shaped structure 111a can be caused to exit lumen 131 of inner shaft 130 — 1008

↓

An interconnect 112 can be caused to exit lumen 131 — 1010

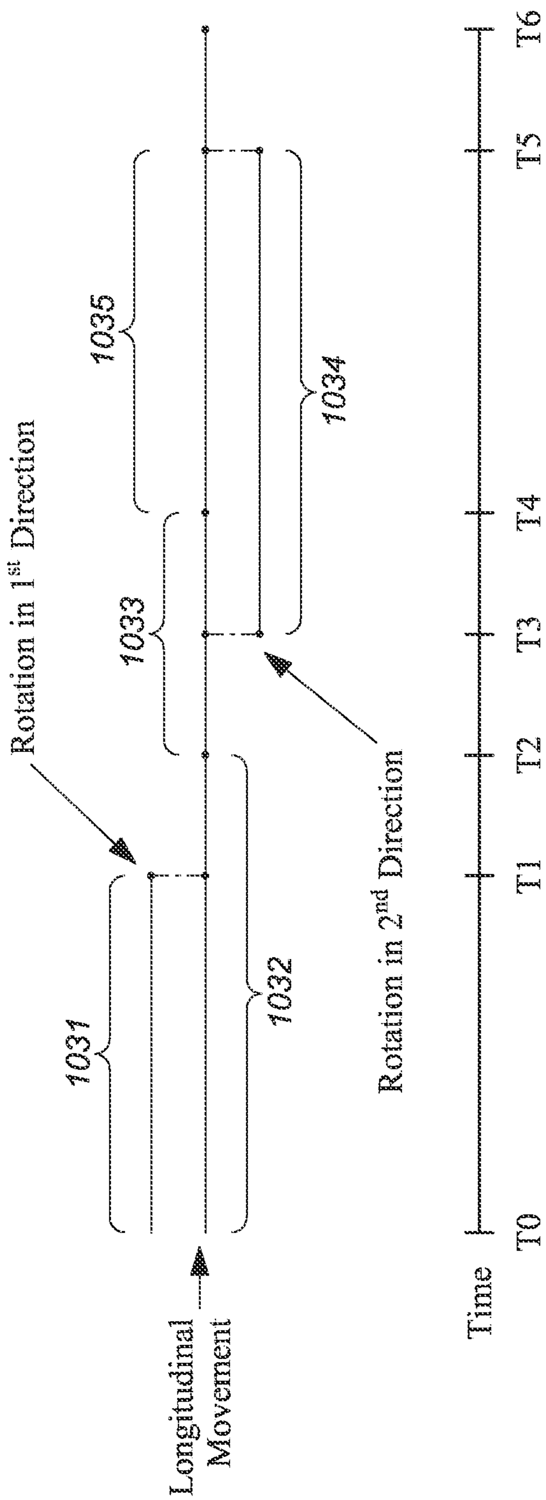
↓

A second ring-shaped structure 111b can be caused to exit lumen 131 — 1012

↓

Distal engagement portion 114 and proximal engagement portion 115 of implant 102 can be released from distal control member 140 and grasper 136, respectively — 1014

FIG. 6A



Rotation in 1st Direction
Rotation in 2nd Direction
1031
1032
1033
1034
1035
Longitudinal Movement
Time
T0
T1
T2
T3
T4
T5
T6

FIG. 6B

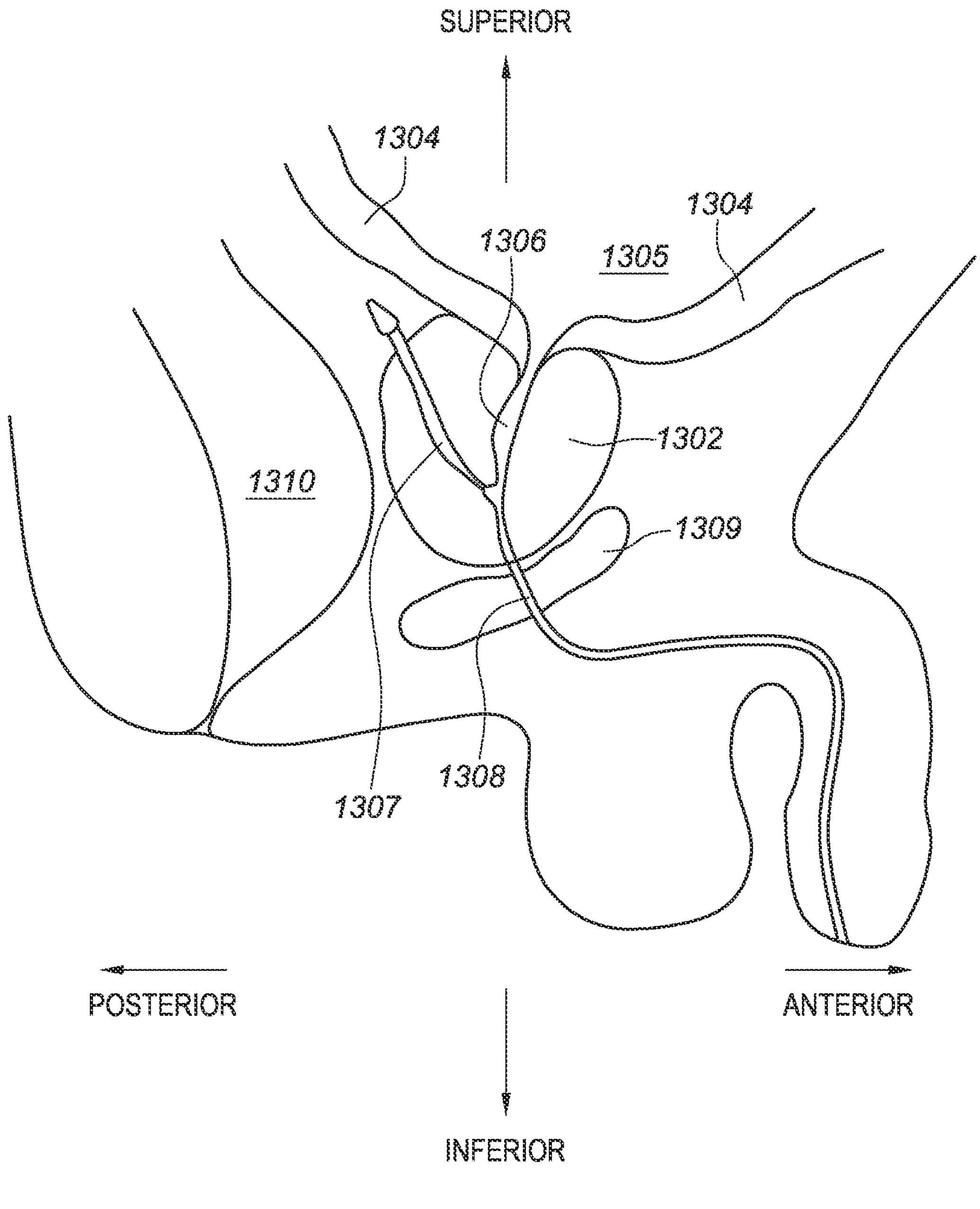

SUPERIOR
1304
1304
1306
1305
1302
1310
1309
1307
1308
POSTERIOR
ANTERIOR
INFERIOR

FIG. 7

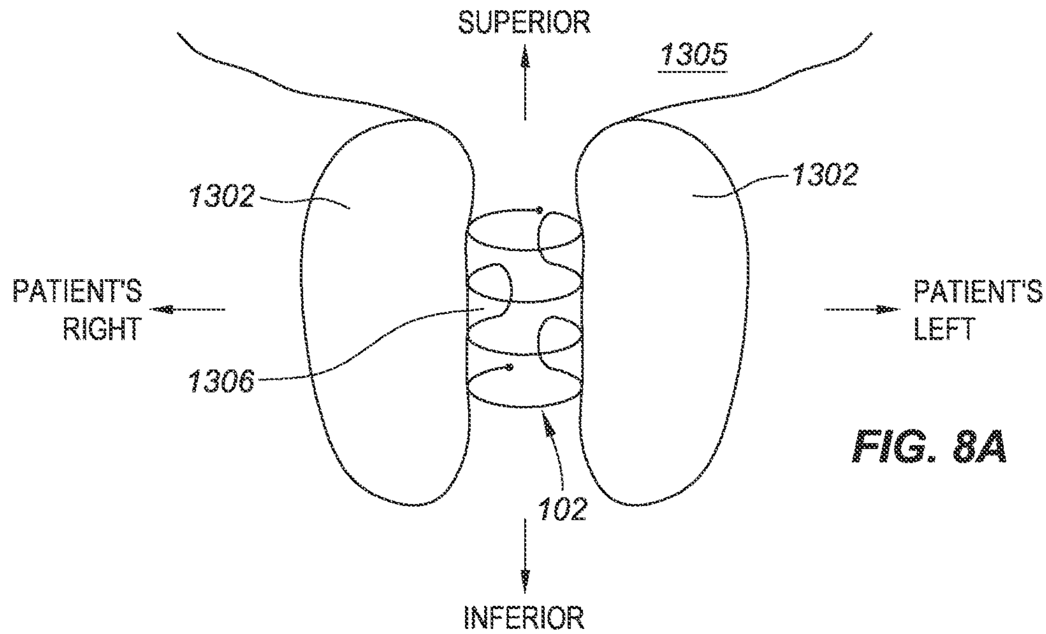


FIG. 8A

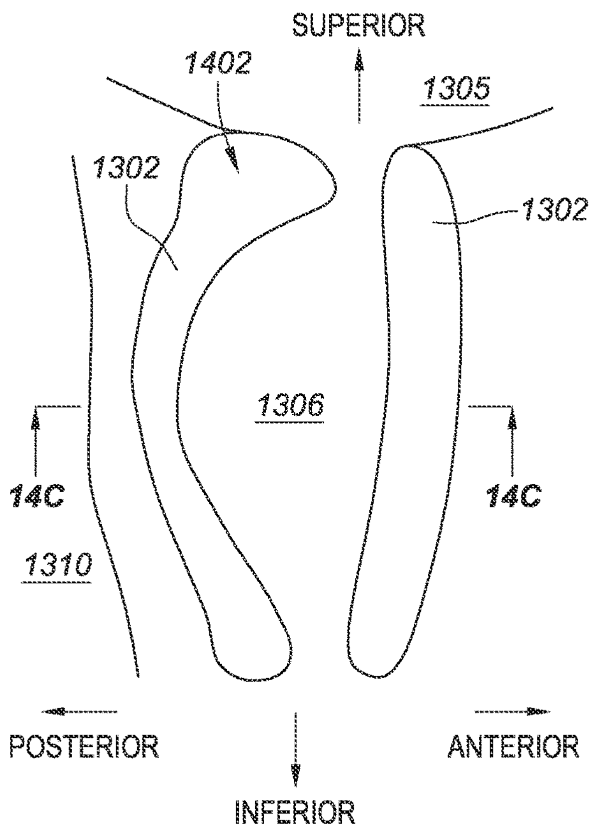


FIG. 8B

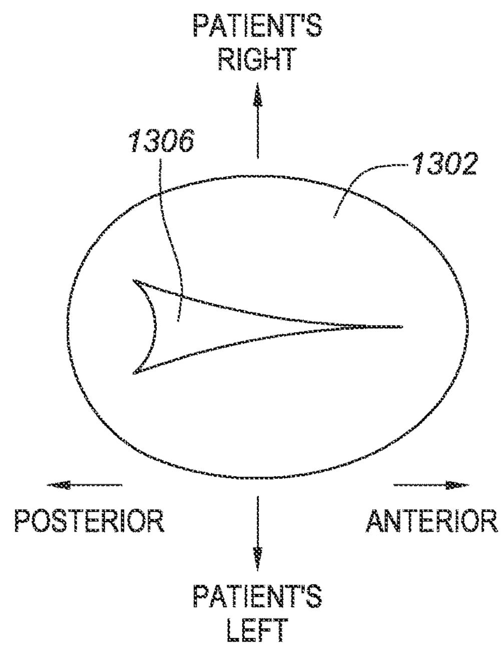


FIG. 8C

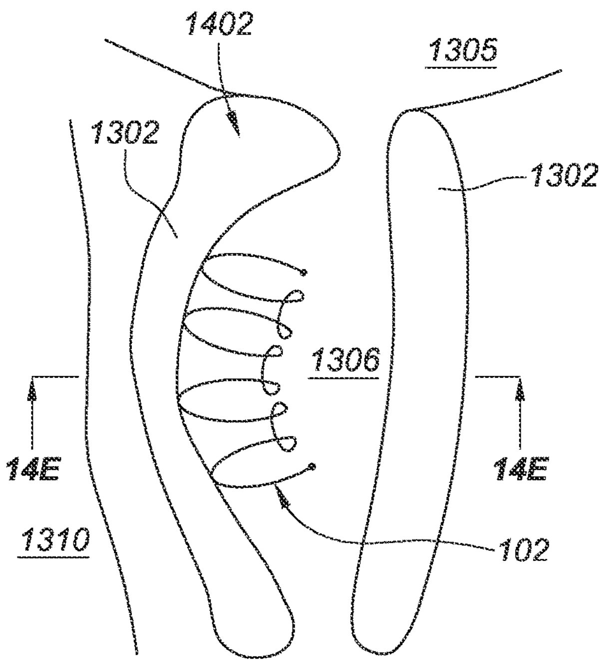


FIG. 8D

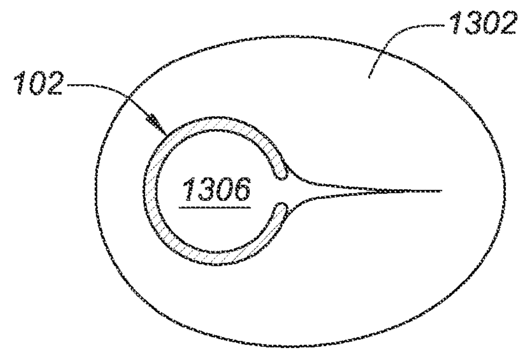


FIG. 8E

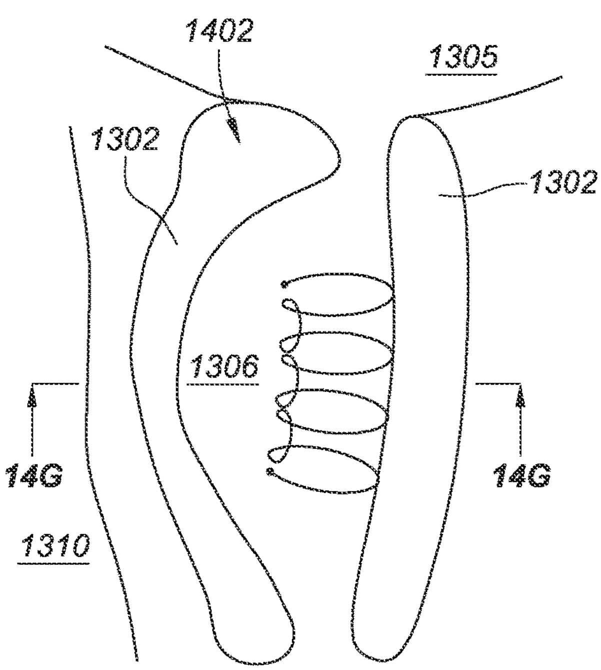


FIG. 8F

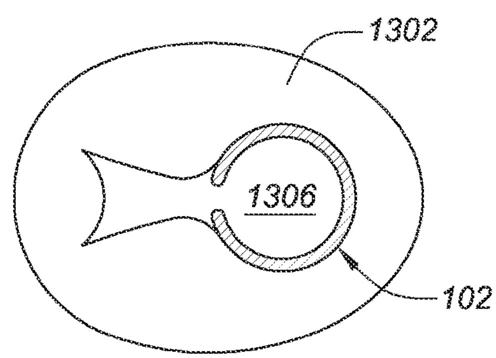


FIG. 8G

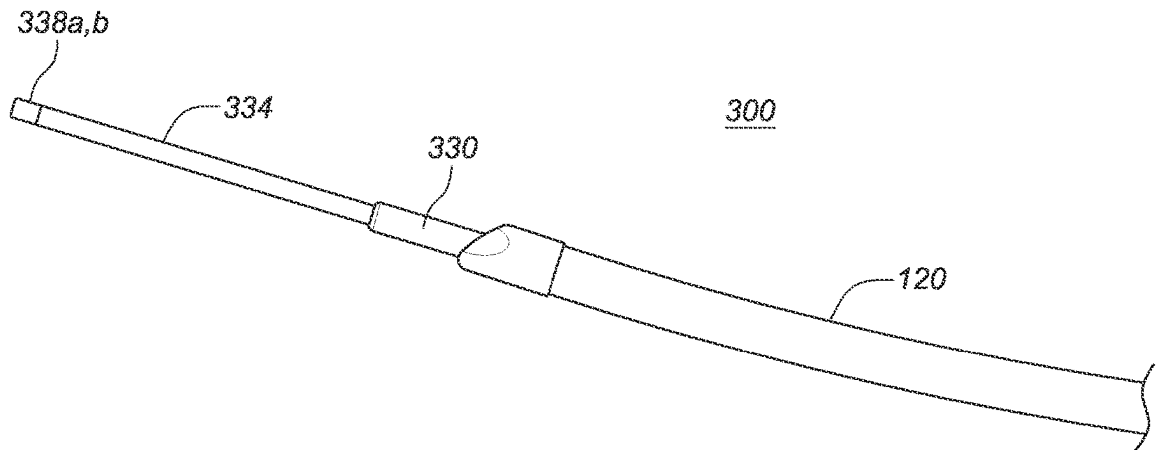


FIG. 9A

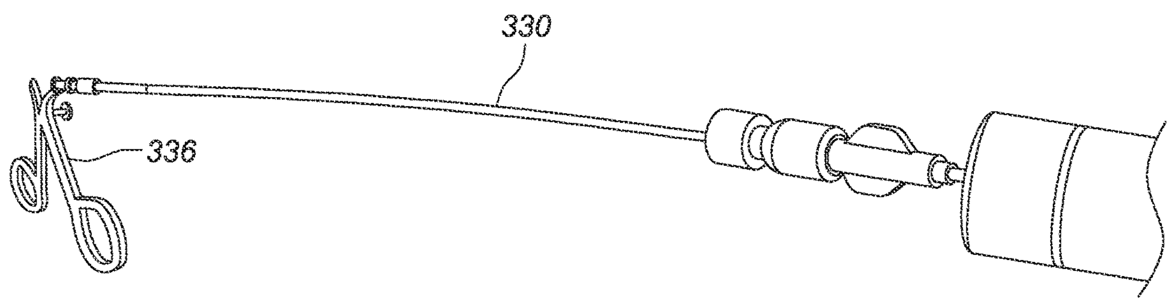


FIG. 9B

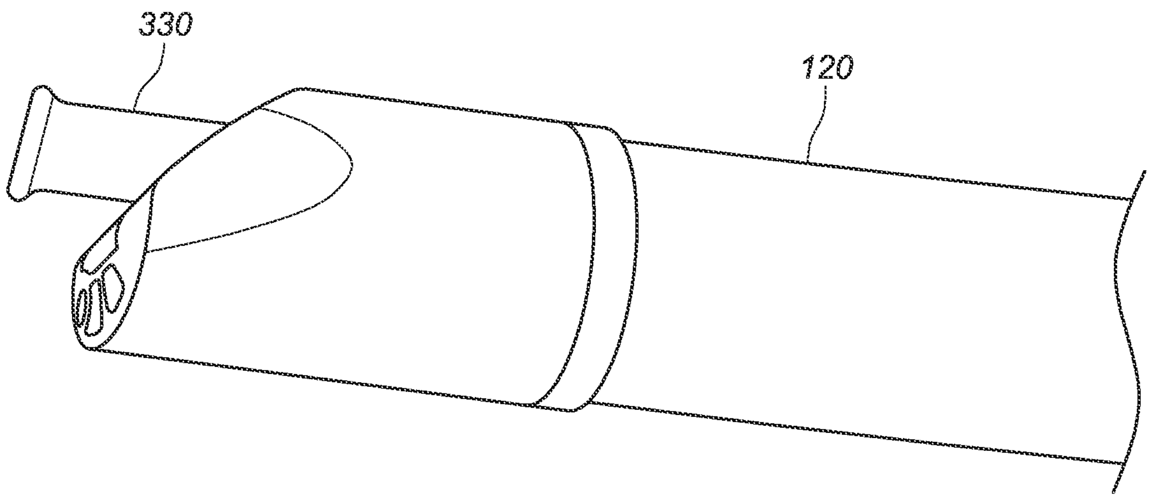


FIG. 9C

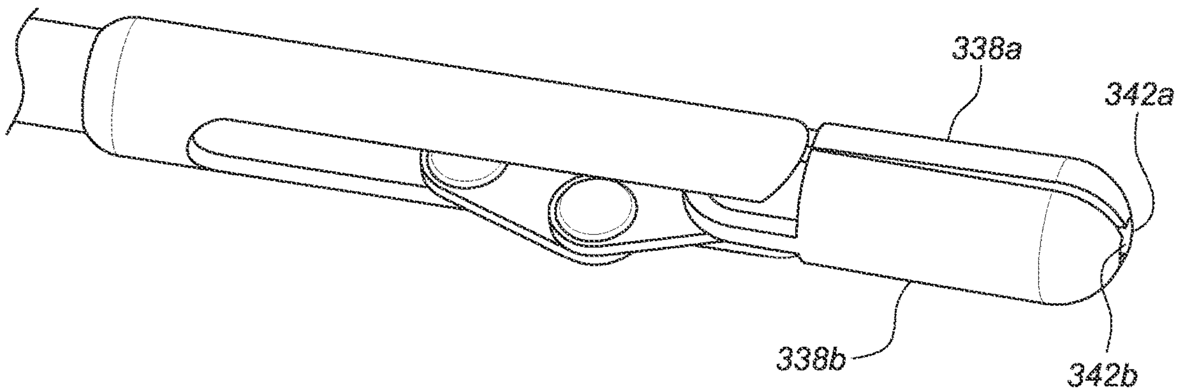


FIG. 10A

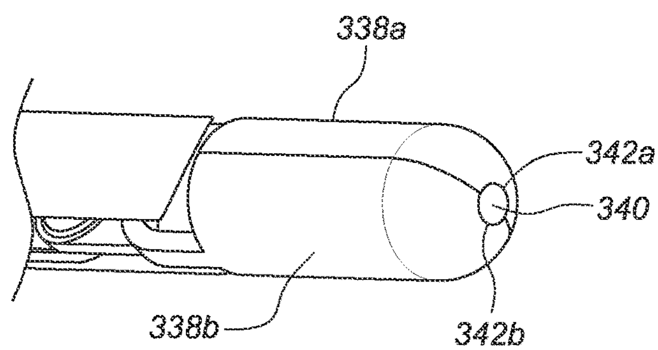


FIG. 10B

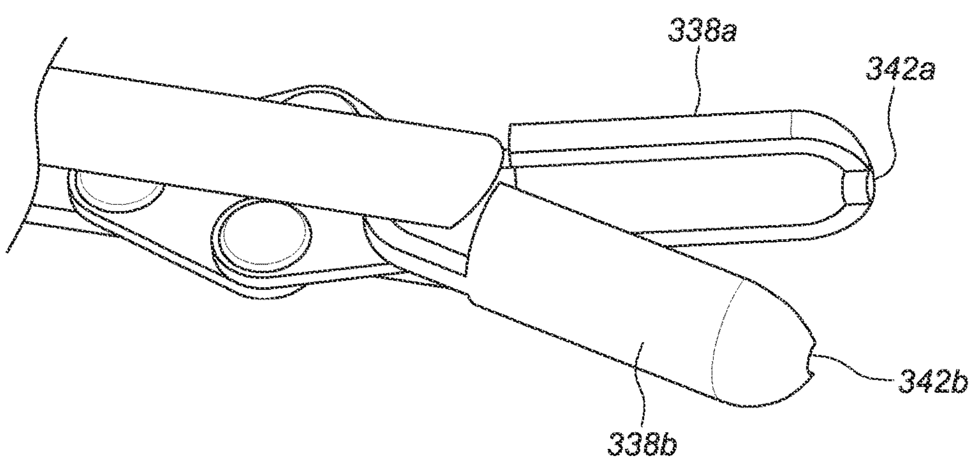


FIG. 10C

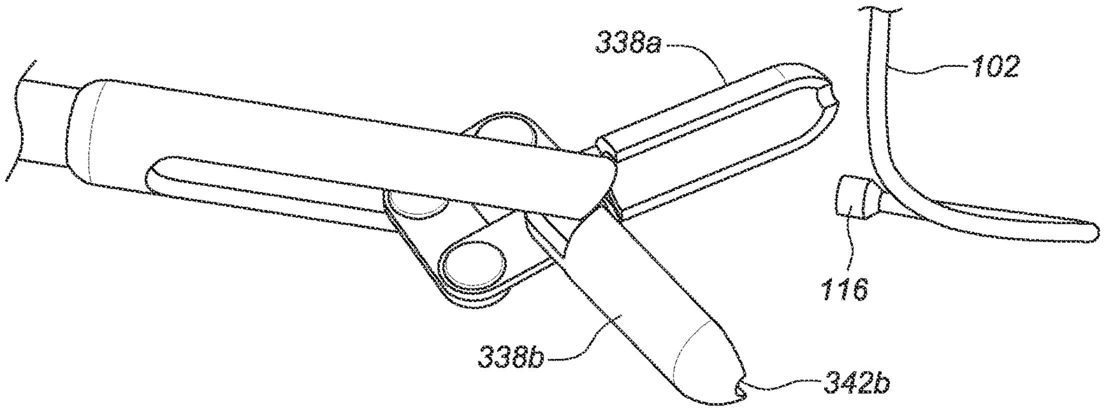


FIG. 10D

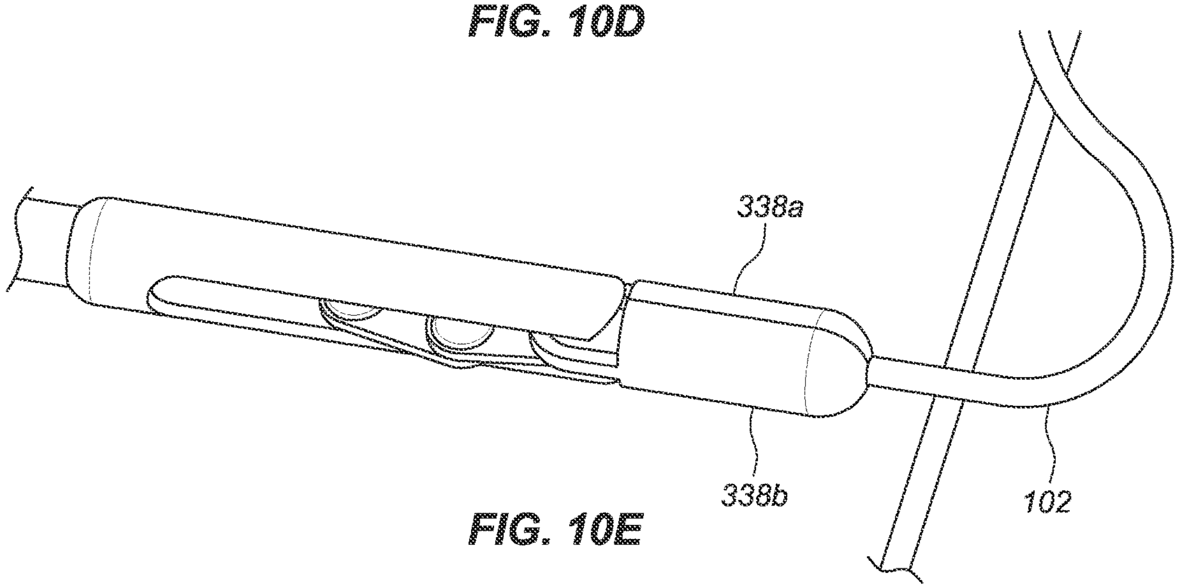


FIG. 10E

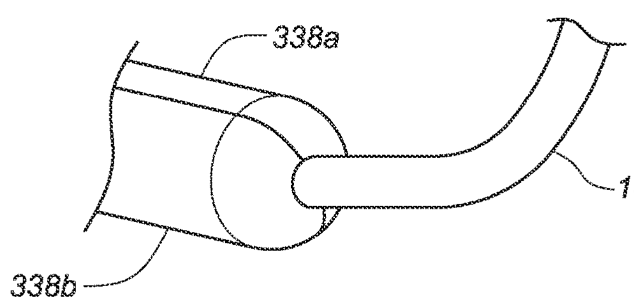


FIG. 10F

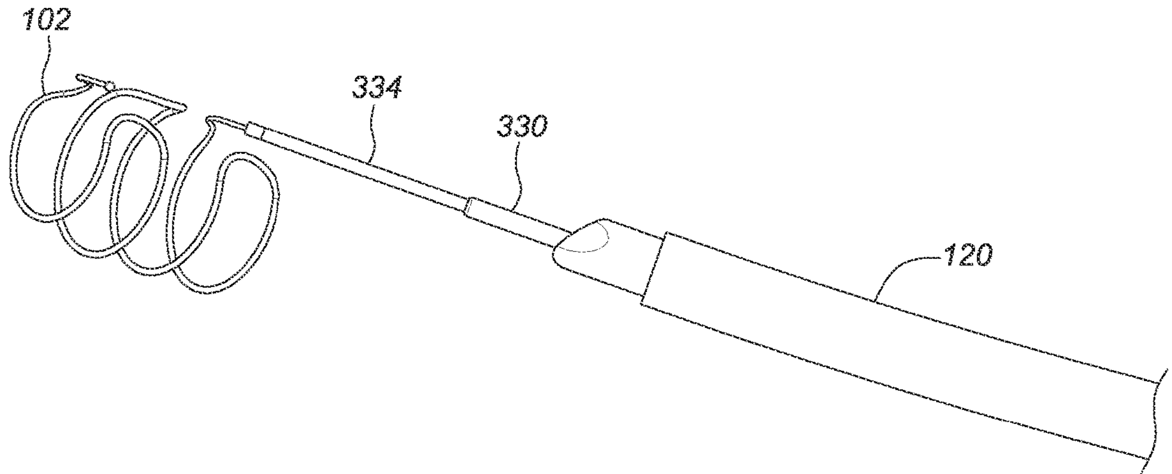


FIG. 11A

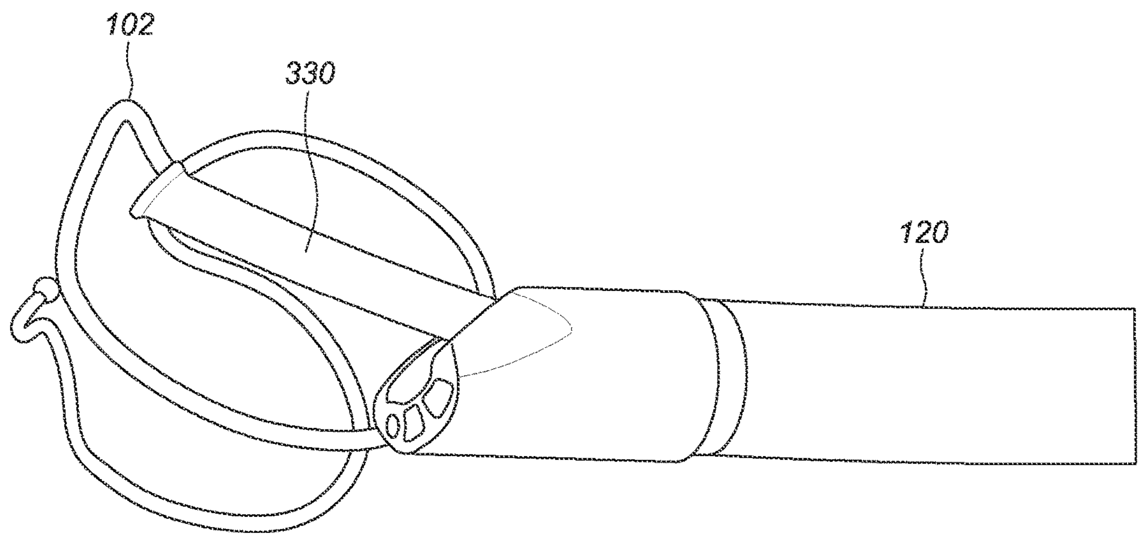


FIG. 11B

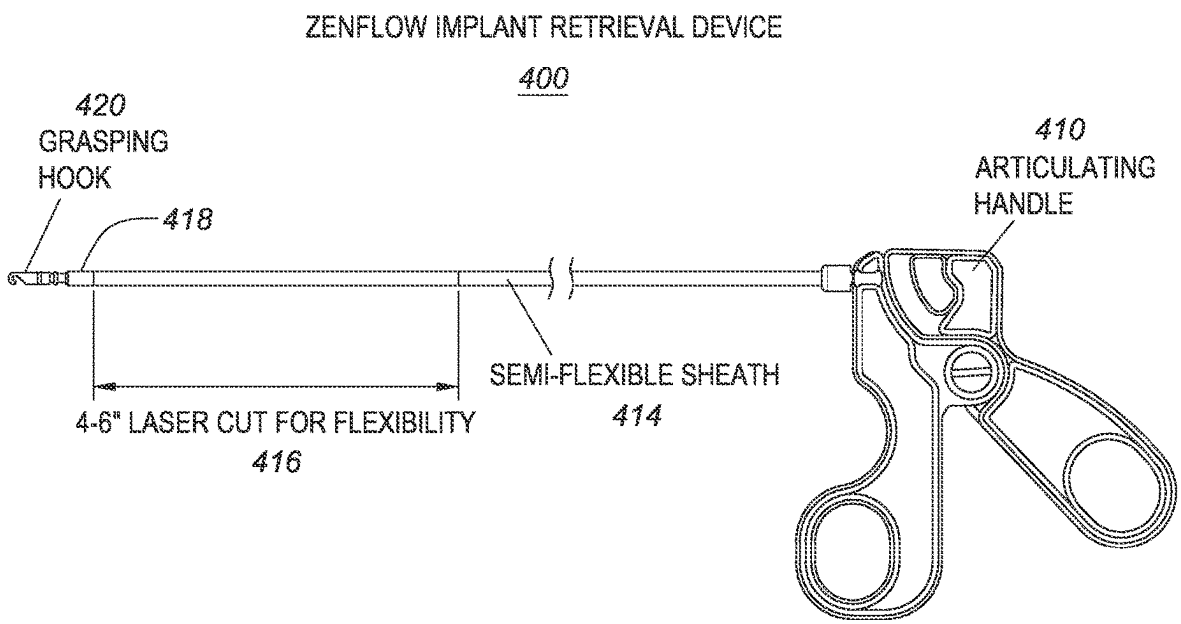

ZENFLOW IMPLANT RETRIEVAL DEVICE
400
420
GRASPING
HOOK
418
410
ARTICULATING
HANDLE
SEMI-FLEXIBLE SHEATH
414
4-6" LASER CUT FOR FLEXIBILITY
416

FIG. 12

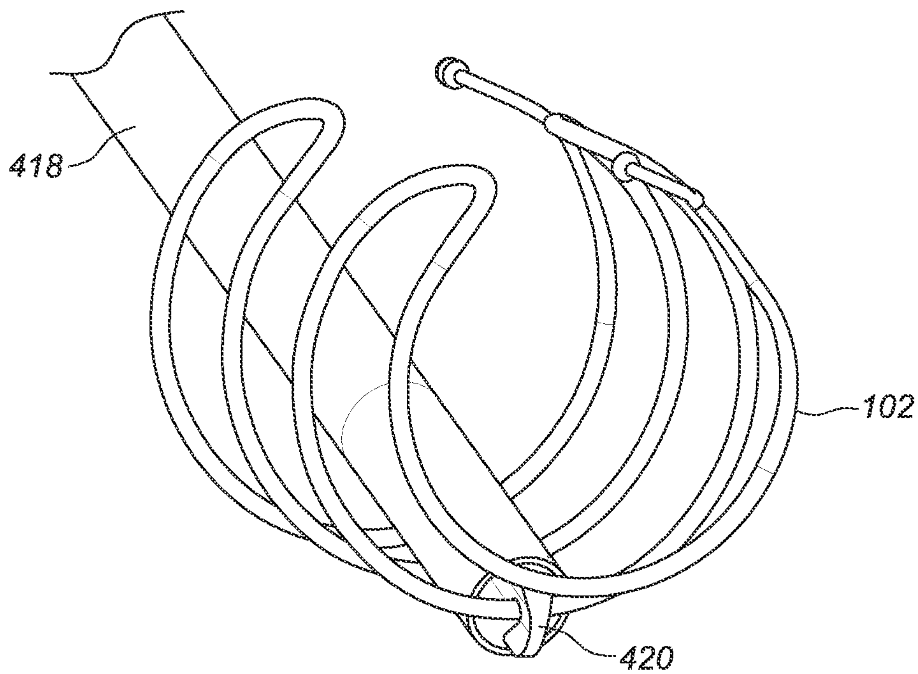


FIG. 13A

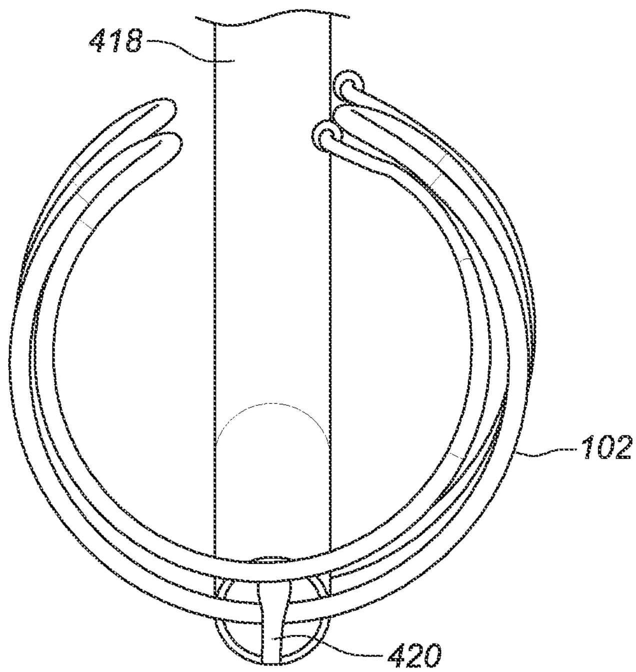


FIG. 13B

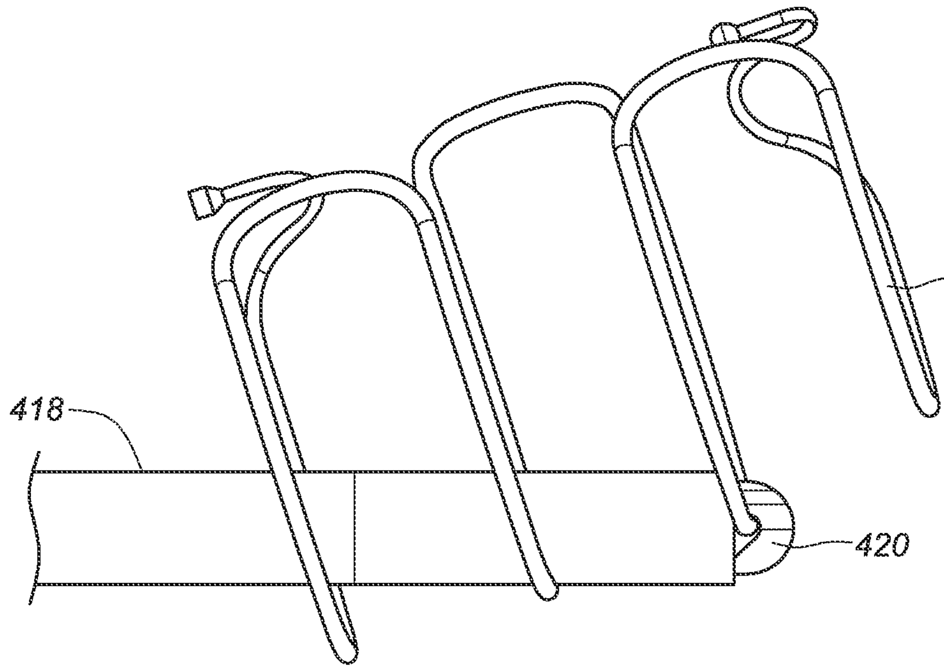


FIG. 13C

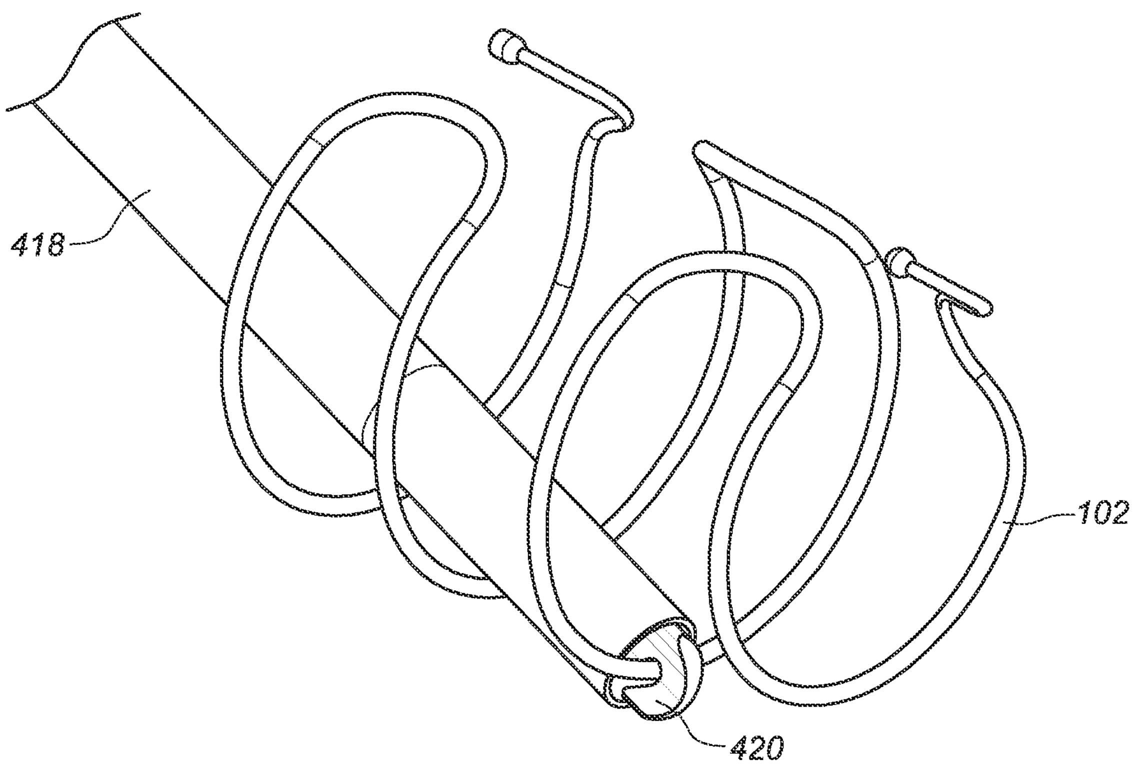


FIG. 13D

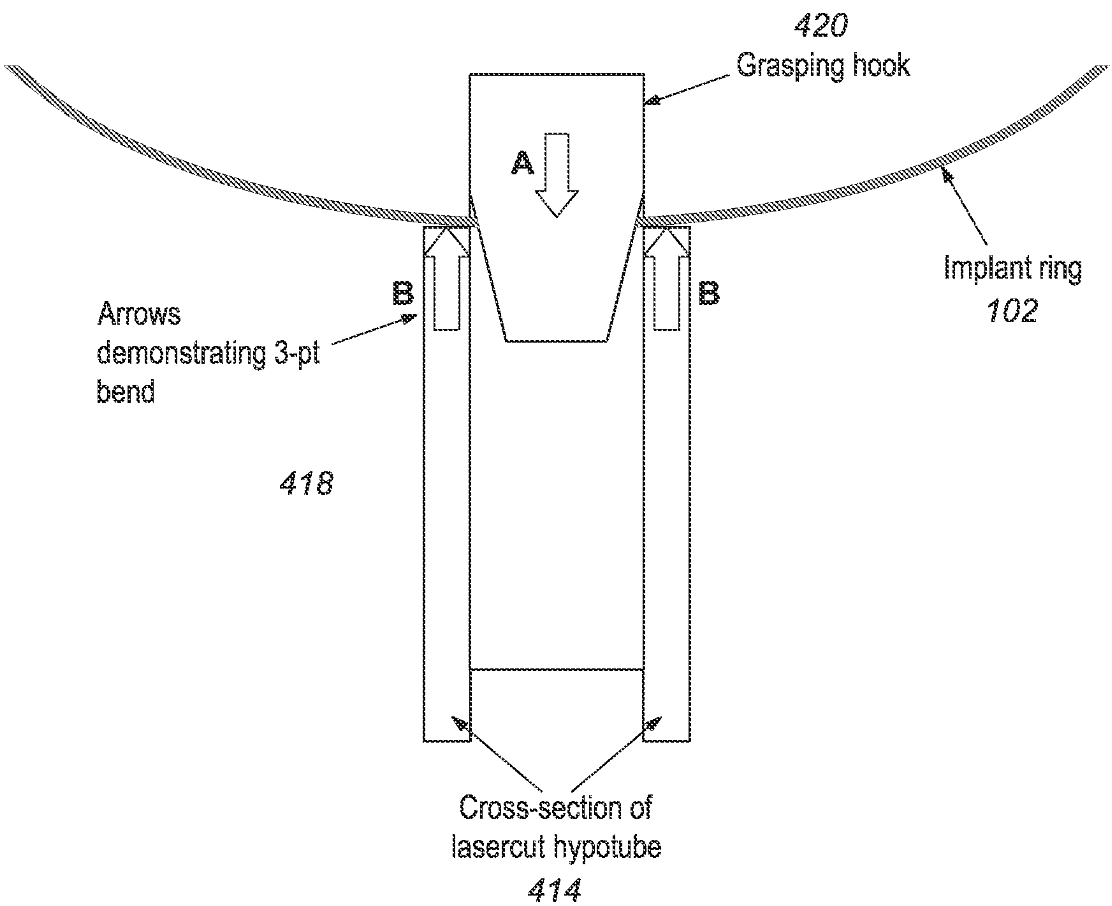

420
Grasping hook
A
Implant ring
102
B
B
Arrows
demonstrating 3-pt
bend
418
Cross-section of
lasercut hypotube
414

FIG. 14

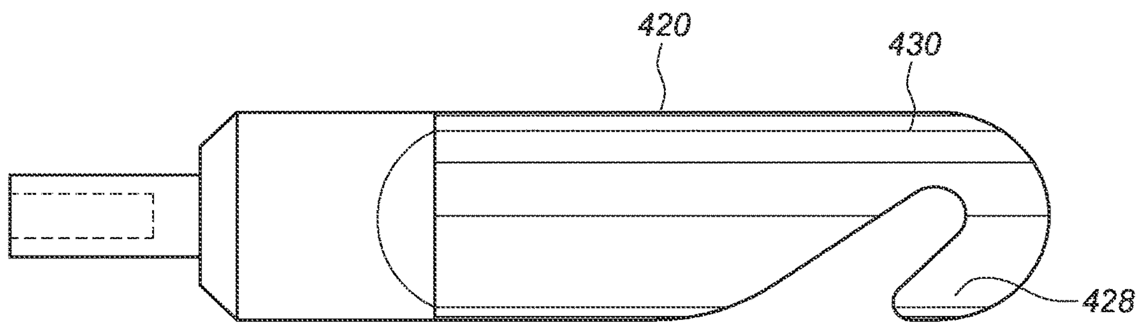


FIG. 15A

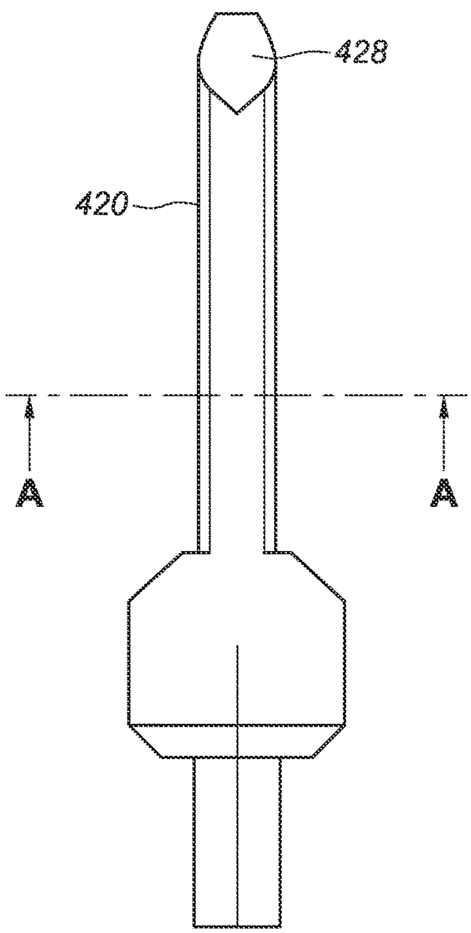


FIG. 15B

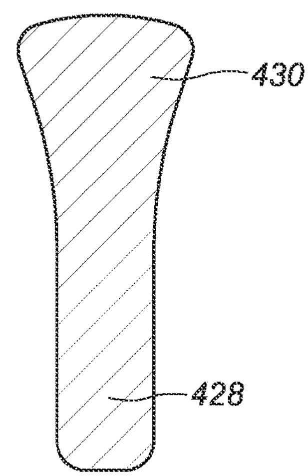


FIG. 15C

SYSTEMS, DEVICES, AND METHODS FOR THE RETRIEVAL OF AN IMPLANT IN THE PROSTATIC URETHRA

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims priority to, and the benefit of, U.S. Provisional Application No. 63/089,205, filed Oct. 8, 2020, which is hereby expressly incorporated by reference in its entirety for all purposes.

STATEMENT OF GOVERNMENT SPONSOR RESEARCH

This invention was made with government support under NIH SBIR Phase II R44DK124094 awarded by the National Institutes of Health. The government has certain rights in the invention.

FIELD

The subject matter described herein relates to systems, devices, and methods for delivery or deployment of an implant into the prostatic urethra, more specifically, delivery in an atraumatic and minimally-invasive manner through the tortuous bends of the male urethra.

BACKGROUND

There are numerous clinical reasons for placement of an implant into the prostatic urethra, such as for treatment of urinary retention associated with benign prostatic hyperplasia (BPH), blockages from prostate cancer, bladder cancer, urinary tract injury, prostatitis, bladder sphincter dyssynergia, benign or malignant urethral stricture, and other conditions for which treatment is desired. Due to the naturally complex and tortuous anatomical geometry, patient-to-patient geometric and tissue variability, and anatomical restrictions associated with those conditions, accurate and consistent placement of an implant into the prostatic urethral lumen has proven challenging. Furthermore, complex challenges are presented in the design and/or fabrication of systems with sufficient flexibility to deliver such an implant in a minimally-invasive manner. For these and other reasons, needs exist for improved systems, devices, and methods of implant delivery to the prostatic urethra.

SUMMARY

Provided herein are a number of example embodiments of delivery systems for delivering or deploying implants within the prostatic urethra or other parts of the body, and methods related thereto. Embodiments of the delivery system can include a delivery device insertable into the prostatic urethra and a proximal control device coupled with the delivery device and configured to control deployment of one or more implants from the delivery device. In some embodiments, the delivery device can include multiple tubular components each having various functions described in more detail herein. Embodiments of the delivery system have imaging capabilities. Multiple embodiments of implants for use with the delivery systems are also described, as are various implanted placements of those implants.

Other systems, devices, methods, features and advantages of the subject matter described herein will be or will become apparent to one with skill in the art upon examination of the following figures and detailed description. It is intended that all such additional systems, methods, features and advantages be included within this description, be within the scope of the subject matter described herein, and be protected by the accompanying claims. In no way should the features of the example embodiments be construed as limiting the appended claims, absent express recitation of those features in the claims.

BRIEF DESCRIPTION OF THE FIGURES

The details of the subject matter set forth herein, both as to its structure and operation, may be apparent by study of the accompanying figures, in which like reference numerals refer to like parts. The components in the figures are not necessarily to scale, emphasis instead being placed upon illustrating the principles of the subject matter. Moreover, all illustrations are intended to convey concepts, where relative sizes, shapes and other detailed attributes may be illustrated schematically rather than literally or precisely.

FIG. **1**A is a block diagram depicting an example embodiment of a delivery system.

FIGS. **1**B, **1**C, and **1**D are side, end, and perspective views, respectively, depicting an example embodiment of an implant.

FIGS. **2**A-**2**B are perspective views depicting example embodiments of a delivery system in different stages of deployment of an implant.

FIGS. **2**C-**2**G are perspective views depicting an example of a release mechanism.

FIGS. **2**H-**2**J are views depicting an alternative example of a release mechanism.

FIGS. **3**A-**3**C are perspective views depicting example embodiments of a grasper component in use within a delivery system.

FIGS. **4**A-**4**C are perspective views depicting an example embodiment of an inner shaft.

FIGS. **4**D-**4**E are cross-sectional views depicting an example embodiment of an inner shaft.

FIGS. **5**A-**5**B are side views depicting an example embodiment of a delivery system in various stages of deployment of an implant.

FIGS. **5**C-**5**F are perspective views depicting an example embodiment of a steering lock device.

FIGS. **5**G-**5**H are cross-sections depicting an example embodiment of a steering lock device.

FIG. **6**A is a flowchart depicting an example embodiment of a method for delivering an implant.

FIG. **6**B is a timing diagram depicting an example embodiment of a sequence of steps for deploying an implant.

FIG. **7** is an example cross-section of the male anatomy.

FIG. **8**A is an example cross-section of the male anatomy having an example embodiment of an implant deployed therein.

FIG. **8**B is an example cross-section of the male anatomy.

FIG. **8**C is an example cross-section of the male anatomy taken along line **8**C-**8**C of FIG. **8**B.

FIG. **8**D is an example cross-section of the male anatomy having an example embodiment of an implant deployed therein and FIG. **8**E is an example cross-section of the male anatomy taken along line **8**E-**8**E of FIG. **8**D.

FIG. **8**F is an example cross-section of the male anatomy having an example embodiment of an implant deployed therein and FIG. **8**G is an example cross-section of the male anatomy taken along line **8**F-**8**F of FIG. **8**G.

FIGS. **9**A-**9**C is an example embodiment of a retrieval device.

FIGS. **10**A-**10**F are example embodiments of a distal end of a retrieval device.

FIGS. **11**A-**11**B are example embodiments of a retrieval device grasping an implant.

FIG. **12** is an example embodiment of an alternative retrieval device.

FIGS. **13**A-**13**D depict perspective views of the distal end of the alternative retrieval device depicted in FIG. **12**.

FIG. **14** is a depiction of forces applied to make a three-point bend.

FIGS. **15**A-**15**B are example embodiments of a hook of a retrieval device.

FIG. **15**C is an example cross-section of a hook of a retrieval device.

DETAILED DESCRIPTION

Before the present subject matter is described in detail, it is to be understood that this disclosure is not limited to the particular embodiments described, as such may, of course, vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting, since the scope of the present disclosure will be limited only by the appended claims.

The subject matter presented herein is described in the context of delivery or deployment of one or more implants within the prostatic urethra. The purpose for deployment of the implant(s) in the prostatic urethra can vary. The embodiments described herein are particularly suited for treatment of BPH, but they are not limited to such. Other conditions for which these embodiments can be used include, but are not limited to, treatment of blockages from prostate cancer, bladder cancer, urinary tract injury, prostatitis, bladder sphincter dyssynergia, and/or benign or malignant urethral stricture. Further, these embodiments can have applicability for deployment of one or more implants in other locations of the urinary tract or in the bladder, as well as other biological lumens, cavities, or spaces, such as the human vasculature, cardiac system, pulmonary system, or gastro-intestinal tract, including locations within the heart, stomach, intestines, liver, spleen, pancreas, and kidney.

The subject matter presented herein further describes methods for removing the implant from the prostatic urethra. Although the implant is intended to be a permanent implant, removal is required under certain circumstances, such as initial implant misplacement, implant migration, safety issues, or efficacy issues. The methods of removal described herein are simple and do not cause any permanent tissue damage. The device can be removed acutely (during a procedure) or chronically (after years of implantation).

FIG. **1**A is a block diagram depicting an example embodiment of delivery system **100** having an elongate delivery device **103** coupled with a proximal control device **200**. A distal end region **104** is adapted to be inserted into the patient's urethra (or other lumen or body cavity of the patient) through the urethral orifice. Distal end region **104** preferably has an atraumatic configuration (e.g., relatively soft and rounded) to minimize irritation or trauma to the patient. Elongate delivery device **103** carries or houses one or more implants **102** (not shown) to be delivered or deployed within or adjacent to the prostatic urethra. A proximal end region **105** of delivery device **103** is coupled with proximal control device **200**, which remains outside of the patient's body and is configured to be used by the physician or other healthcare professional to control the delivery of one or more implants **102**.

Example Embodiments of Delivery Devices and Related Methods

FIGS. **1**B, **1**C, and **1**D are side, end, and perspective views, respectively, depicting an example embodiment of implant **102** in an at-rest configuration. Implantable device **102** is biased towards the at-rest configuration depicted here and is deformable between the at-rest configuration and a relatively more elongate housed (or delivery) configuration (e.g., see FIG. **3**A) for housing implant **102** within delivery device **103**. The housed configuration can be a straight or lineated state with minimal curvature. The at-rest configuration has a relatively greater lateral width, and a relatively shorter longitudinal length than the housed configuration. Upon exiting an open end of delivery device **103**, implant **102** is free to transition its shape back towards that of the at-rest configuration although restraints imparted by the patient's urethral wall may prevent implant **102** from fully reaching the at-rest configuration. Because implant **102** is biased towards the at-rest configuration, implant **102** is configured to automatically expand when freed from the restraint of delivery device **103**, and can be referred to as "self-expanding." The shape of implant **102** in its deployed state within, e.g., the patient's urethra, can be referred to as the deployed configuration, and will often be a shape that is deformed from the at-rest configuration by the surrounding tissue, although the deployed configuration can be the same as the at-rest configuration.

Implant **102** can be configured in numerous different ways, including any and all of those implant configurations described in U.S. Patent Publ. No. 2015/0257908 and/or Int'l Publ. No. WO 2017/184887, both of which are incorporated by reference herein for all purposes.

Implant **102** can be formed from one or more discrete bodies (e.g., wires, ribbons, tubular members) of varying geometries. Referring to the embodiment of FIGS. **1**B-**1**D, implant **102** has a main body formed of only one single wire member set in a predetermined shape. Implant **102** can have two or more ring-shaped structures **111** (in this embodiment there are four: **111**a, **111**b, **111**c, and **111**d) with one or more interconnections **112** extending between each pair of adjacent ring-shaped structures **111** (in this embodiment there is one interconnection between each adjacent pair, for a total of three: **112**a, **112**b, and **112**c). Each interconnection **112** extends from one ring-shaped structure **111** to an immediately adjacent ring-shaped structure **111**. Each interconnection **112** can have a relatively straight shape (not shown) or a curved (e.g., semi-circular or semi-elliptical) shape as shown in FIGS. **1**B-**1**D.

Ring-shaped structures **111** are configured to maintain the urethra in a fully or partially open state when expanded from the housed configuration. Device **100** can be manufactured in various sizes as desired, such that the width (e.g., diameter) of each ring-shaped structure **111** is slightly larger than the width of the urethra, and the length of each interconnection **112** determines the spacing between ring-shaped structures **111**. Ring-shaped structures **111** can have the same or different widths. For example, in the embodiment depicted here, ring-shaped structure **111**a has a relatively smaller width than structures **111**b-**111**d, which have the same width. This can accommodate prostatic urethras that converge to a smaller geometry before the bladder neck.

Each ring-shaped structure **111** can be located or lie in a single plane, and in some embodiments that single plane can be oriented with a normal axis perpendicular to a central axis **124** of implant **102** (as depicted in FIG. **1**B). In other embodiments, ring-shaped structures **111** can be located in multiple planes. Ring-shaped structures **111** can extend around central axis **126** to form a complete circle (e.g., a 360 degree revolution) or can form less than a complete circle (e.g., less than 360 degrees) as shown here. Although not limited to such, in many embodiments ring-shaped structures **111** extend between 270 and 360 degrees.

As can be seen from FIGS. **1B-1D**, the geometry of implant **102** can have a cylindrical or substantially cylindrical outline shape with a circular or elliptical cross-section. In other embodiments, implant **102** can have a prismatic or substantially prismatic shape with triangular or substantially triangular cross-section, or otherwise.

Implant **102** can also include a distal engagement member **114** and a proximal engagement member **115** that are each configured to engage with elements of delivery device **103**. Engagement with delivery device **103** can serve one or more purposes such as allowing control of the release of implant **102**, allowing movement of the ends of implant **102** relative to each other, and/or allowing retrieval of implant **102** after deployment, e.g., in an instance where the physician desires to recapture implant **102** and redeploy implant **102** in a different position. In this embodiment, distal engagement member **114** is a wire-like extension from ring-shaped structure **111***a* that has a curved (e.g., S-like) shape for positioning an atraumatic end **116** (e.g., rounded, spherical, ballized) in a location suitable for engagement with delivery device **103** and thereby allow control of the distal end region of implant **102**. Likewise, proximal engagement member **115** has a curved shape for positioning another atraumatic end **117** in a location suitable for engagement with delivery device **103** and thereby allow control of the proximal end region of implant **102**. In other embodiments, distal engagement member **114** and proximal engagement member **115** can be configured such that the atraumatic ends **116** and **117** point in different directions. For example, atraumatic ends **116** and **117** can be pointing distally instead of proximally. In another embodiment, atraumatic ends **116** and **117** can be pointing in opposite directions (e.g., atraumatic end **116** can be pointing distally and atraumatic end **117** can be pointing proximally, and vice versa). In other embodiments, distal engagement member **114** and proximal engagement member **115** can be omitted, and delivery device **103** can couple with implant **102** at one or more other distal and/or proximal locations, such as on a ring-shaped structure **111** or interconnect **112**. Moreover, an extension having an atraumatic end (similar to distal engagement member **114** and proximal engagement member **115**) can be attached in the middle of implant **102** in order to provide an additional structure to control placement of the middle portion of the implant.

Delivery device **103** can include one or more elongate flexible members (e.g., **120**, **130**, **140**, and **150** as described below), each having one or more inner lumens. Alternatively, one or more elongate flexible members of delivery device **103** can be a solid or a non-hollow member with no inner lumen. FIG. **2A** is a perspective view depicting an example embodiment of distal end region **104** of a delivery device **103**. In this embodiment, delivery device **103** includes a first elongate tubular member **120**, a second elongate tubular member **130**, a third elongate tubular member **140**, and a fourth elongate tubular member **150**. Delivery device **103** can vary and in other embodiments can include more or less tubular members.

In this embodiment, first elongate tubular member **120** is the outermost tubular member and is flexible yet provides support for members contained therein. First tubular member **120** is referred to herein as outer shaft **120** and can have one or more inner lumens. In this embodiment, outer shaft **120** includes a first inner lumen **121** housing second elongate tubular member **130**, which is referred to herein as inner shaft **130**. Outer shaft **120** and inner shaft **130** are each controllable independent of the other. Inner shaft **130** can slide distally and proximally within lumen **121** and is shown here partially extending from an open distal terminus of outer shaft **120**.

In this embodiment, outer shaft **120** includes three additional lumens **122**, **123**, and **124**. An illumination device (not shown) and an imaging device (not shown) can be housed in two of lumens **122-124** (e.g., lumens **122** and **123**). The imaging device can utilize any desired type of imaging modality, such as optical or ultrasound imaging. In one example embodiment the imaging device utilizes a forward (distal) looking CMOS imager. The illumination device can be configured to provide adequate illumination for optical imaging, and in one embodiment includes one or more light emitting diodes (LEDs). In embodiments where illumination is not required, such as for ultrasound imaging, the illumination device and its respective lumen can be omitted or the lumen could be used for an alternative purpose, e.g., as an irrigation or flushing channel. The illumination device and/or the imaging device can each be fixedly secured at the distal terminuses of lumens **122** and **123**, or each can be slidable within lumens **122** and **123** to allow advancement further distally from outer shaft **120** and/or retraction into outer shaft **120**. In one example embodiment, the illumination device and the imaging device are mounted together and only a single lumen **122** or **123** is present for that purpose. The remaining lumen (e.g., lumen **124**) can be configured as an irrigation or flush port from which fluid such as saline can be introduced to the urethra to flush the region and provide adequate fluid through which implant **102** and the surrounding prostatic urethra wall can be imaged. In one embodiment, the outer shaft may contain two separate lumens for fluid management. One lumen may be used for irrigation and the other lumen may be used for flushing.

Outer shaft **120** has a proximal end (not shown) coupled with proximal control device **200**. Delivery device **103** can be configured to be steerable to navigate tortuous anatomy. Steerability can be unidirectional (e.g., using a single pull wire) or multidirectional (e.g., using two or more pull wires arranged at different radial locations about device **103**) depending on the needs of the application. In some embodiments, the structures (e.g., pull wires) for steerability extend from distal end region **104** of delivery device **103** (e.g., where the distal ends of the pull wires are secured to a plate or other structure within distal end region **104**) to proximal control device **200**, where they can be manipulated by the user to steer delivery device **103**. The steering structures can be located in one or more lumens of outer shaft **120**, or can be coupled to or embedded within a sidewall of outer shaft **120**. Delivery device **103** can be biased to deflect in a particular lateral direction (e.g., bend) such that device **103** automatically deflects in that manner and forces imparted to steer delivery device **103** are in opposition to this biased deflection. Other mechanisms for steering delivery device **103** can also be used. The steering mechanism may also be locked or adjusted during deployment of implant **102** to control the position of implant **102** within the anatomy (e.g., steering anteriorly during deployment may help place implant **102** in a more desirable anterior position).

Inner shaft **130** can include one or more inner lumens for housing one or more implants **102** and/or other components. In this embodiment, inner shaft **130** includes a first lumen **131** in which one or more implants **102** can be housed, and a second lumen **132** in which third elongate tubular member **140** can be housed. In this embodiment, third elongate tubular member **140** is configured to releasably couple with the distal end region of implant **102** and is referred to as a distal control member or tether **140**. Distal control member **140** can be slidably advanced and/or retracted with respect to inner shaft **130**. Distal control member **140** can include an inner lumen **141** that houses fourth elongate tubular member **150**, which is shown here extending from an open distal terminus of distal control member **140**. Fourth elongate tubular member **150** is configured to anchor delivery device **103** with respect to the patient's anatomy, e.g., to keep components of delivery device **103** stationary with respect to the anatomy during deployment of implant **102**, and is referred to as anchor delivery member **150**.

In the configuration depicted in FIG. **2**A, anchor delivery member **150** is extended from lumen **141** of distal control member **140**, and distal control member **140** along with inner shaft **130** are shown extended from lumen **121** of outer shaft **120**. When delivery device **130** is advanced through the urethra, anchor delivery member **150** is preferably housed entirely within distal control member **140**, and distal control member **140** along with inner shaft **130** are retracted from the positions shown in FIG. **2**A such that they reside within lumen **121** of outer shaft **120** and do not extend from the open distal terminus of lumen **120**. In other words, in some embodiments the open distal terminus of outer shaft **120** forms the distalmost structure of device **103** upon initial advancement through the urethra. This facilitates steering of delivery device **103** by outer shaft **120**. The physician can advance distal end region **104** of delivery device **103** to be in proximity with the desired implantation site, or entirely into the patient's bladder. Anchor delivery member **150** can be exposed from the open distal terminus of distal control member **140**, either by distally advancing anchor delivery member **150** further into the bladder, or if already present within the bladder, then by proximally retracting the other components of delivery device **103**. At this point the anchor from anchor delivery member **150** can be deployed in the bladder.

FIG. **2**B is a perspective view depicting distal end region **104** of delivery device **103** with the various components deployed. In this embodiment, anchor delivery member **150** includes an anchor **152** in the form of an inflatable member or balloon.

Other embodiments of anchors **152** are described in International Application No. PCT/US19/32637, filed May 16, 2019, which is hereby incorporated by reference in its entirety for all purposes. Anchor **152** expands (or otherwise transitions) to a size greater than that of the bladder neck such that anchor **152** resists proximal retraction (e.g., a relatively light tension). In embodiments where anchor **152** is a balloon, that balloon can be an elastic or inelastic and inflatable with an inflation medium (e.g., air or liquid such as saline) introduced into balloon **152** through one or more inflation ports **153**. Here three inflation ports **153** are located on the shaft of anchor delivery member **150** and communicate with an inflation lumen that extends proximally back to proximal control device **200**, which can include a port for inflation with a syringe. Upon deployment of anchor **152**, the physician can proximally retract delivery system **100** until anchor **152** is in contact with the bladder neck and/or wall (if not already).

The physician can use the imaging device of outer shaft **120** to move delivery device **103** proximally away from anchor **152** until the physician is in the desired position within the urethra to begin deployment of implant **102**. A retainer **142** on distal control member **140** is releasably coupled with distal engagement member **114** of implant **102**. The physician can position retainer **142** in a location along the length of the urethra where the physician desires the distal end of implant **102** to deploy. This can involve moving distal control member **140** and inner shaft **130**, together, proximally and/or distally with respect to anchor delivery member **150**. In another embodiment, the position of retainer **142** is fixed with respect to anchor **152** such that the longitudinal position of implant **102** within the anatomy is set by the system independently of any manipulation by the physician. The coupling of distal engagement member **114** with retainer **142** also permits the physician to manipulate the radial orientation of implant **102** by rotating distal control member **140** and inner shaft **130** together. Active or passive shaping of distal control member **140** may allow for a more desirable placement of implant **102**. For example, member **140** may have a curvature that places the implant in a more anterior anatomical position. This curvature may be inherently set in member **150** or actively applied by the physician though a separate entity such as a control wire. Once in the desired location and orientation, the physician can proximally retract inner shaft **130** with respect to distal control member **140** to initiate deployment of implant **102**.

Distal engagement member **114** is held in place with respect to distal control member **140** by retainer **142**, and proximal retraction of inner shaft **130** with respect to distal control member **140** causes ring-shaped structures **111** to begin to deploy in sequence (**111***a*, then **111***b*, then **111***c*, then **111***d* (not shown)). Distal control member **140** can remain stationary or be moved longitudinally with respect to the urethra during deployment. In some embodiments, distal control member **140** is steerable to allow for angulation of implant **102** to accommodate relatively tortuous anatomy. The steerability of distal control member **140** can also accomplish relatively anterior placement of the implant relative to the bladder neck, which potentially contributes to improved flow results. For example, see distal control member **140** as shown in FIGS. **2**C-**2**G and FIGS. **10**C and **10**D. Mechanisms for accomplishing steerability are discussed elsewhere herein and can likewise be applied to distal control member **140**. In these or other embodiments, distal control member **140** can be significantly flexible to passively accommodate tortuous anatomy. In some embodiments, distal control member **140** has a predefined curve to assist in navigation.

To assist in deployment, inner shaft **130** can rotate clockwise and counterclockwise (as depicted by arrow **134**) about distal control member **140**. Referring back to FIGS. **1**B-**1**C, implant **102** has a non-constant direction of winding that, when viewed as commencing at distal engagement member **114**, proceeds clockwise along ring-shaped structure **111***a*, then reverses along interconnect **112***a* to a counterclockwise direction for ring-shaped structure **111***b*, then reverses along interconnect **112***b* to a clockwise direction for ring-shaped structure **111***c*, and then reverses along interconnect **112***c* to a counterclockwise direction for ring-shaped structure **111***d*, until ending at proximal engagement member **115**. Depending on the direction of winding of the portion of implant **102** about to exit the open distal terminus of lumen **131**, the transition of implant **102** towards the at-rest configuration can impart a torque on shaft **130** if shaft **130** is not actively rotated as implant **102** is deployed. That torque can cause shaft **130** to passively rotate (without user intervention) either clockwise or counterclockwise accordingly. In certain embodiments described elsewhere herein, shaft **130** is actively rotated during deployment. Rotation of inner shaft **130** with respect to distal control member **140** thus allows delivery device **103** to rotate and follow the direction of winding of implant **102**. In some embodiments, all ring-shaped structures **111** are wound in the same direction, clockwise or counterclockwise (e.g., as in the case of a fully spiral or helical implant), or do not have a set direction of winding.

In this or other embodiments, the distal end region of inner shaft **130** is configured to be relatively more flexible than the more proximal portion of inner shaft **130**, which can permit avoidance of excessive motion of the rest of device **103** during deployment, resulting in better visualization and less tissue contact by device **103**. Such a configuration can also reduce the stress imparted on implant **102** by device **103** during delivery. For example, the portion of inner shaft **130** extending from outer shaft **120** during deployment can be relatively more flexible than the portion of inner shaft **130** that remains within outer shaft **120**, thus allowing inner shaft **130** to flex more readily as implant **102** exits inner lumen **131**. This in turn can stabilize delivery device **103** and allow the physician to obtain stable images of the appointment process.

In an alternative embodiment, as seen in FIGS. **4**A-**4**E, inner shaft **230** can include an outer torqueing tube **233** (FIGS. **4**B-**4**E), one or more lumens for housing one or more implants **102** and/or other components, and one or more torqueing supports **235**. In this embodiment, inner shaft **230** includes a first elongate tubular member **231***a* having a first lumen **231** in which one or more implants **102** can be housed. First elongate tubular member **231***a* also has a second elongate tubular member **232***a* (or tether) having a second lumen **232** in which a third elongate tubular member **140** and a fourth elongate tubular member **240**, which could act as an inflation lumen, can be housed. In an alternative embodiment, the second elongate tubular member **232***a* (or tether) can be used for release/actuation and the inflation lumen can be concentric with the tether. As seen in FIGS. **4**D and **4**E, the first **231***a* and second **232***a* elongate tubular members can sit side-by-side and be held in place by the torqueing supports **235**. The torqueing supports **235** can be small plates spaced within the outer torqueing tube **233** from a proximal to a distal end of outer torqueing tube **233**. For example, the torqueing supports **235** may be placed about 3 to about 6 inches apart, alternatively about 2 to about 5 inches apart, alternatively about 1 to about 4 inches apart. The torqueing supports **235** can be bonded or otherwise fixed in place relative to the outer torqueing tube **233** to ensure that axial and angular position of the outer torqueing tube **233** can be maintained by the user. The first elongate tubular member **231***a* can be fixed to the torqueing supports **235** to ensure that the first elongate tubular member **231***a* moves with the outer torqueing tube **233**. The second elongate tubular member **232***a* may not be fixed to the torqueing supports **235** so that the second elongate tubular member **232***a* can move axially and rotationally relative to the support plate and outer torqueing tube **233**.

As seen in FIG. **4**B, the flexible tip **243** may be created by fixing the first elongate tubular member or implant delivery tube **231***a* such that its distal end **237** extends beyond the distal tip **239** of the outer torqueing tube **233** by between about 0 cm and 1.5 cm, alternatively between about 0 cm and 1.0 cm, and alternatively between about 0.2 and 1.0 cm.

The components of the inner shaft may be made from appropriate materials. The first elongate tubular member or implant delivery tube **231***a* may be a braided tubular assembly with a lubricious liner. It may be made from a laser cut hypotube with a lubricious liner, a single polymer extrusion, or other appropriate material. The outer torqueing tube **233** may be made from a laser cut hypotube, a braided construction, a polymer extrusion, or other appropriate material. The torqueing supports **235** may be laser-cut metal plates, molded plastic components, extruded materials, or other appropriate material.

FIG. **2**B depicts implant **102** after three ring-shaped structures **111***a*, **111***b*, and **111***c* have been deployed. Proximal retraction of shaft **130** continues until the entirety of implant **102**, or at least all of ring-shaped structures **111**, have exited lumen **131**. If the physician is satisfied with the deployed position of implant **102** and the deployed shape of implant **102**, then implant **102** can be released from delivery device **103**. A control wire **146** (not shown in FIG. **2**B) extends within the length of control member **140**, either in the same lumen as anchor delivery member **150** or in a different lumen, and is coupled to retainer **142**. Control wire **146** can be routed into member **140** through an opening **148**.

Release of the distal end of implant **102** can be accomplished by releasing retainer **142**. Retainer **142** can be a cylindrical structure or other sleeve that linearly or rotationally actuates over a cavity or recess in which a portion of implant **102** is housed. In the embodiment of FIG. **2**B, retainer **142** includes an opening or slot that allows distal engagement member **114** to pass therethrough. Retainer **142** can rotate with respect to the cavity or recess in which distal engagement member **114** (not shown) is housed until the opening or slot is positioned over member **114**, at which point member **114** is free to release from distal control member **130**. Rotation of retainer **142** can be accomplished by rotation of a rotatable shaft, rod or other member coupled with retainer **142** (and accessible at proximal control device **200**). Alternative embodiments of retainers can be found in FIGS. 2C-2F of International Application No. PCT/US19/32637, filed May 16, 2019, which was previously incorporated by reference in its entirety for all purposes.

FIGS. **2**C-**2**G are perspective views depicting another example embodiment of system **100** with an alternative retainer **142** that can be fixed in position with a tether lock. As in other embodiments, retainer **142** slides distally and/or proximally with respect to distal control member **140**. Distal engagement member **114** of implant **102** can be received within a corresponding recess **143** (FIG. **2**G) of distal control member **140**. Retainer **142** can slide over distal engagement member **114** while received within this recess **143** until retainer **142** abuts a portion of member **140**, which has opening **241** located near its distal end. A control wire **246** extends within the length of control member **140**, either in the same lumen as anchor delivery member **150** or in a different lumen, and attaches or couples to retainer **142** at its distal end **248**. As seen in FIG. **2**E, control wire **246** passes out of and back into opening **241** in distal control member **140**, such that control member **246** forms a loop **247** that protrudes from the opening and extends along an axis perpendicular to a longitudinal axis of the distal control member and a longitudinal axis of retainer **142**. Loop **247**, which is located adjacent to and proximal of retainer **142**, prevents retainer **142** from moving in a proximal direction over distal control member **140**.

Upon satisfactory deployment of implant **102** within the urethra, e.g., in the state of FIG. **2**C, control wire **246** can be tensioned by pulling control wire **246** in a proximal direction (away from the implant **102**). As seen in FIG. **2**F, the tension pulls loop **247** into the lumen of distal control member **140**, thereby removing the obstruction preventing retainer **142** from sliding proximally. After the loop is withdrawn into the lumen of distal control member **140**, as seen in FIG. **2**G, retainer **140** is proximally retracted by further pulling control wire **246** proximally to expose engagement member **114** and permit its release from member **140**.

Control member **146**, **246** may be made from nitinol, Kevlar, stainless steel, suture, liquid crystal polymers (LCP) or any other tensionable material.

FIGS. **2**H-**2**J illustrate another example embodiment of system **100** with an alternative retainer **242** that can be fixed in position. As with other embodiments described, retainer **242** can be a cylindrical structure or other sleeve that linearly or rotationally actuates over a cavity or recess in which a portion of implant **102** is housed. Retainer **242** includes cover **245** that is coupled to an outer tube **249** that extends to the control device **200**. In the embodiment of FIGS. **2**H-**2**J, retainer **242** includes an opening or slot (not shown) that allows distal engagement member **114** to pass therethrough. FIG. **2**H shows cover **245** closed over the recess **143** that is adapted to hold distal engagement member **114**. Retainer **242** can be withdrawn proximally with respect to the cavity or recess in which distal engagement member **114** is housed until the opening or slot is positioned over member **114**, at which point member **114** is free to release from distal control member **130**. As seen in FIG. **2**I, cover **245** has been withdrawn by actuating outer tube **249** proximally. Withdrawal of cover **245** of retainer **242** can be accomplished by withdrawing outer tube **249** proximally, which is accessible at proximal control device **200**. FIG. **2**J is a cross-section showing the retainer **242** and the inflation lumen that communicates with anchor **152**. The inflated diameter of the anchor balloon can be between about 1 cm and 7 cm, alternatively between about 2 cm and 6 cm, alternatively between about 1 cm and 6 cm.

Release of the proximal end of implant **102** is also controllable. FIG. **3**A is a partial cross-sectional view depicting an example embodiment of system **100** with a portion of implant **102** shown within inner lumen **131** of inner shaft **130**. Here, implant **102** is in the lineated state prior to deployment with proximal engagement member **115** coupled with a grasper **136** that is slidable distally and/or proximally within lumen **131**. Grasper **136** can include a distal end region **137** on or coupled with a shaft **138**. Grasper **136** is preferably controllable to rotate and longitudinally translate (e.g., push and pull) implant **102** with respect to inner shaft **130**.

FIGS. **3**B and **3**C are perspective views depicting an example embodiment of distal end region **137** of grasper **136** without implant **102** and with implant **102**, respectively. Grasper **136** includes a recess (also referred to as a cavity or pocket) **139** for receiving and holding proximal engagement member **115**. Here, the enlarged portion **115** is retained within recess **139** by a distal necked down region having a relatively smaller width. While within inner lumen **131**, the sidewalls of inner shaft **130** maintain proximal engagement member **115** within recess **139**. When distal end region **137** exits inner lumen **131** (either by retracting inner shaft **130** with respect to grasper **136** or by advancing grasper **136** with respect to inner shaft **130**), the restraint imparted by the inner shaft sidewalls is no longer present and engagement member **115** is free to release from grasper **136**. Thus, when the physician is satisfied with placement of the deployed implant **102**, distal engagement member **114** can be released by moving retainer **142** and permitting distal engagement member **114** to decouple from control member **140**, and proximal engagement member **115** can be released by exposing grasper **136** from within inner shaft **130** and permitting proximal engagement member **115** to decouple from grasper **136**.

Grasper **136** can also assist in loading implant **102**. In some embodiments, application of a tensile force on implant **102** with grasper **136** (while the opposite end of implant **102** is secured, for example, by retainer **142**) facilitates the transition of implant **102** from the at-rest configuration to a lineated configuration suitable for insertion of implant **102** into inner shaft **130**.

Anchor delivery member **150** can have multiple different configurations and geometries (e.g., including those that extend in one direction across the bladder wall, two directions across the bladder wall (e.g., left and right), or three or more directions across the bladder wall). Additional examples of anchor delivery members and anchors are described in FIGS. **2**B and **4**A-**4**J of International Application No. PCT/US19/32637, filed May 16, 2019, which was previously incorporated by reference in its entirety for all purposes.

Upon completion of the implant deployment procedure, anchor **152** can be collapsed or retracted to permit removal of delivery device **103**. For instance, in embodiments where anchor **152** is a balloon, that balloon is deflated and optionally retracted back into a lumen of device **103**, and subsequently withdrawn from the bladder and urethra. In embodiments where anchor **152** is a wire form or other expandable member (such as those described with respect to FIGS. **4**A-**4**G of International Application No. PCT/US19/32637, filed May 16, 2019, which was previously incorporated by reference in its entirety for all purposes), anchor **152** is retracted back into the lumen of device **103** from which it was deployed, and device **103** can subsequently be withdrawn from the bladder and urethra. Retraction can be accomplished using fluid or pneumatic actuation, a screw type mechanism, or others.

Example Embodiments of Proximal Control Devices and Related Methods

FIG. **5**A is a side view depicting an example embodiment of delivery system **100** prior to deployment of implant **102**, and FIG. **5**B is a side view depicting this embodiment with implant **102** in a deployed configuration (anchor delivery member **150** and distal control member **140** are not shown). In this embodiment proximal control device **200** is a handheld device having a handle **201**, a first user actuator **202** (configured in this example as a trigger), a main body **203**, and a second user actuator **205**. A longitudinal axis of delivery device **103** is indicated by dashed line **204**. Proximal control device **200** can include mechanisms that are manually powered by actuation of actuator **202** to cause relative motions of the components of device **103**. In other embodiments, proximal control device **200** can utilize electrically powered mechanisms instead. Second user actuator **205** can be configured to control steering of delivery device **103**. Here, as seen in FIGS. **5**G and **5**H, actuator **205** is configured as a rotatable wheel **225** that can wind or unwind a pull wire **221** within delivery device **103** and cause deflection of device **103** upwards and downwards as depicted here. Second user actuator **205** includes an extension **212** having paddle **206** extending from a first end **215** of the extension **212**. As seen in FIG. **5**A, prior to deployment, the extension **212** is closer to handle **201**, e.g., extension **212** is angled toward handle **201**. As seen in FIG. **5**B, after implant **102** has been at least partially deployed from distal end region **104**, extension **212** is angled away from handle **201** and angled or pointed towards distal end region **104**. The dotted lines in FIG. **5**B also indicate that the distal end of the inner tubular member **120** can be deflected to enable placement of the implant further anteriorly. Proximal control device **200** can be configured so that, after all of ring-shaped structures **111** have been deployed from inner lumen **131** but prior to advancement of proximal engagement feature **115** and recess **139** from within lumen **131**, further deployment of implant **102** is automatically prevented. This provides the physician with an opportunity to verify that implant **102** has been properly deployed and placed prior to releasing implant **102** from delivery device **103**. A detailed description of the control device **200** and the parts and gear assemblies contained therein, can be found in, e.g., FIGS. 6A-9F, of International Application No. PCT/US19/32637, filed May 16, 2019, which was previously incorporated by reference in its entirety for all purposes.

The device may also include a steering lock that enables the user to lock the steering anteriorly to place the implant in a more anterior position. As discussed previously, the steerability of the device can include a pull wire **225** that extends from distal end region **104** of delivery device **103** (e.g., where the distal ends of the pull wires are secured to a plate or other structure within distal end region **104**) to proximal control device **200**, where they can be manipulated by the user to steer delivery device **103**. The steering structures can be located in one or more lumens of outer shaft **120**, or can be coupled to or embedded within a sidewall of outer shaft **120**. Delivery device **103** can be biased to deflect in a particular lateral direction (e.g., bend) such that device **103** automatically deflects in that manner and forces imparted to steer delivery device **103** are in opposition to this biased deflection.

The steering lock is part of extension **212** attached to actuator **205**. As seen in FIGS. **5**C-**5**H, actuator **205** includes a rotatable wheel **225**, an extension **212**, a latch **209**, and a ledge **207**. The housing of actuator **205** may include two halves, a right handle half **205***a* and a left handle half **205***b*. The rotatable wheel **225** is adapted to wind and unwind the pull wire and is located in and coupled to the housing. Extension **212** includes latch **209** and paddle **206**, which extends from a first end **215** and terminates in detent **208**, such that a gap exists between detent **208** and a second end **217** of extension **212**. The second end **217** of extension **212** is attached to left handle half **205***b* and the first end is adjacent a portion of the right handle half **205***a*. The second end **217** of extension **212** includes the detent **208** and gap. The steering lock also includes ledge **207** that extends from right handle half **205***a* of the housing in proximity to the first end **215** of extension **212**. Latch **209** is adapted to actuate or slide along the paddle **206**. When latch **209** is located on the second end **217**, detent **208** frictionally engages latch **209**, thereby restraining latch **209** to the second end **217**.

In use, as seen in FIG. **5**E, the user can disengage latch **209** from detent **208** and move latch **209** along paddle **206** from the second end **217** to the first end **215** of extension **212**. Once latch **209** is at the first end **215**, extension **212** can be pushed in a direction towards distal end region **104** by the user until latch **209** comes into contact with ledge **207**. Ledge **207** then frictionally engages latch **209** and holds extension **212** in a position angled toward distal end region **104** in a "locked" position. In the locked position, the rotatable wheel **225** cannot wind or unwind the pull wire **221** and the user cannot move (deflect or straighten) the distal end region **104** of the outer tubular member **103**. As seen in FIG. **5**F, to release paddle **206** from the "locked" position, the user can release latch **209** from ledge **207** and slide latch **209** along paddle **206** from the first end **215** to the second end **217** of extension **212**. When latch **209** is no longer frictionally engaged by ledge **207**, extension **212** can passively return to a rest position in which extension **212** is angled toward handle **201** (i.e., away from distal end region **104**) due to spring-loading. In the unlocked position, the rotatable wheel **225** is capable of winding and unwinding the pull wire **221**, thereby moving (deflecting or straightening) the distal end region **104** of the outer tubular member **103**.

Example Embodiments of Delivery Methods

FIG. **6**A is a flow diagram depicting an example embodiment of a method **1000** of delivering implant **102** using system **100**. Distal end region of outer shaft **120** is inserted into the urethra, preferably with inner shaft **130**, distal control member **140**, and anchor delivery member **150** in retracted states fully contained within outer shaft **120** such that no part is extending from the open distal terminus of outer shaft **120**. After advancement into the urethra, at step **1002** anchor delivery member **150** is advanced distally with respect to the remainder of delivery device **103** (e.g., members **120**, **130**, and **140**) and used to deploy anchor **152** within the bladder. In some embodiments, deployment of anchor **152** can be the inflation of one or more balloons (e.g., as depicting in FIG. **2**B,) by the introduction of an inflation medium through an injection (e.g., luer taper) port. The longitudinal positioning (e.g., advancement and retraction) of anchor delivery member **150** and/or any wire-form members can be accomplished manually by the user manipulating a proximal end of anchor delivery member **150** and/or any wire-form members either directly or with proximal control device **200**.

At step **1004**, anchor **152** can be held in tension against the bladder wall by exertion of a proximally directed force on device **200**. Anchor **152** can therefore provide an ordinate for system **100** from which to deploy implant **102** in an accurate location. This feature can ensure the implant is not placed too close to the bladder neck.

At **1006**, distal control member **140** and inner shaft **130** can then be distally advanced from within outer shaft **120** if they have not already (for example, step **1006** can occur prior to steps **1002** and/or **1004**). The user can manipulate the position of proximal control device **200** with the aid of imaging (as described herein) until implant **102** is in the desired position. Once implant **102** is in the desired position, the implant deployment procedure can begin. The steps for implant deployment can be performed automatically by user actuation of proximal control device **200** (e.g., actuation of trigger **202**, selection of a position for switch **604**, etc.), or the steps can be performed directly by hand manipulation of each component of delivery device **103**, or by a combination of the two as desired for the particular implementation.

In some embodiments, deployment of implant **102** from within lumen **131** is fully accomplished by (1) distally advancing grasper **136** with respect to inner shaft **130**, while inner shaft **130** is not moved, while in other embodiments, deployment of implant **102** from within inner lumen **131** is fully accomplished by (2) proximally retracting inner shaft **130** with respect to grasper **136** while grasper **136** is not moved. In some embodiments, deployment of implant **102** is fully accomplished by (3) a combination of both movements. In still other embodiments, deployment of implant **102** is fully accomplished by (1), (2), or (3) in combination with one or more rotations of inner shaft **130**, in one or more directions (e.g., clockwise or counterclockwise) with respect to distal control member **140**.

An example embodiment of a sequence of steps **1008**, **1010**, and **1012** for deploying implant **102** is described with reference to FIG. **6**A and the timing diagram of FIG. **6**B. First with reference to FIG. **6**A, at step **1008** a first ring-shaped structure **111***a* is caused to exit lumen **131** of inner shaft **130**, at step **1010** an interconnect **112** is caused to exit lumen **131**, and at step **1012** a second ring-shaped structure

111*b* is caused to exit lumen **131**. Steps **1010** and **1012** can be repeated for each additional interconnect **112** and ring-shaped structure **111** present on implant **102**.

In FIG. **6**B, step **1008** begins at the far left of the timing diagram at T**0**. Deployment of ring-shaped structure **111***a* corresponds to the duration of time marked **1008**, deployment of interconnect **123** corresponds to time span **1010**, and deployment of ring-shaped structure **111***b* corresponds to time span **1012**. Those of ordinary skill in the art will recognize that the differentiations between deployment of a ring-shaped structure **111** and deployment of an interconnect **112** are approximations as the transitions between those portions of implant **102** can be gradual and do not have to have precise demarcations.

The embodiment described with respect to FIG. **6**B is for an implant with ring-shaped structures **111** having opposite directions of winding (e.g., clockwise, then counterclockwise, then clockwise, etc.). Three different motions are indicated in FIG. **6**B. At top is rotational motion of inner shaft **130** in one direction (e.g., clockwise), in the middle is longitudinal motion (e.g., proximal or distal) of one or more components of delivery device **103**, and at bottom is rotational motion inner shaft **130** in the direction opposite (e.g., counterclockwise) that indicated at top. In embodiments where ring-shaped structures **111** of implant **102** are all wound in the same one direction, rotation of inner shaft **130** will also be in only one direction.

From time T**0** to T**1**, deployment of implant **102** is accomplished by rotating inner shaft **130**, as indicated in region **1031**. At the same time, in region **1032**, grasper **136**, and thus implant **102**, is distally advanced without moving outer shaft **120** longitudinally (neither distally nor proximally) nor rotationally, and also without longitudinally moving inner shaft **130** (neither distally nor proximally).

From time T**1** to T**2**, rotation of inner shaft **130** is stopped but distal advancement of grasper **136** continues while shafts **120** and **130** do not move longitudinally.

From time T**2** to T**4**, deployment of a first interconnect **112** takes place. In region **1033**, from time T**2** to T**4**, no distal advancement of grasper **136** (and implant **102**) occurs. Deployment of interconnect **112** is accomplished by proximal retraction of both outer shaft **120** and inner shaft **130** while holding grasper **136** in place. This causes interconnect **112** to exit inner lumen **131** of shaft **130**.

With respect to rotation of inner shaft **130**, from time T**2** to T**3** no rotation of inner shaft **130** occurs. Within proximal control device **200** the interrupted portion of annular gear **802** continues and there is no rotation of shaft **130** by central gear **816**.

In embodiments where interconnect **112** is straight, then it can be desirable to refrain from rotating shaft **130** while interconnect **112** is deployed from time T**2** to T**4**. For embodiments where interconnect **112** is curved, such as the embodiment of FIGS. **1**B-**1**D, it may be desirable to initiate rotation of inner shaft **130** during interconnect deployment. FIG. **6**B depicts deployment for a curved interconnect **112**, and from T**3**-T**4** inner shaft **130** is rotated in the opposite direction as indicated by region **1034**.

At T**4**, deployment of interconnect **112** is complete and deployment of second ring-shaped structure **111***b* begins. Proximal retraction of shafts **120** and **130** is stopped as indicated by the cessation of region **1033**. Distal advancement of grasper shaft **138** is restarted in region **1035** at T**4**, while outer shaft **120** is not moved rotationally nor longitudinally. Rotation of inner shaft **130** continues as indicated in region **1034**, but inner shaft **130** is not moved longitudinally

These motions continue until time T**5**, at which point rotation of inner shaft **130** is stopped. Within proximal control device **200**, an interrupted portion of annular gear **802** is reached and gear **802** disengages from the planetary gears and rotation of central gear **816** is stopped. User depression of trigger **202** continues from time T**5**-T**6**, the components operate with similar motions as described from time T**1** to T**2**. If another interconnect **112** and ring-shaped structure **111** are present, then the sequence beginning at time T**6** can be the same as that described beginning at time T**2** and continuing to time T**6**.

In many embodiments described here, deployment of all of ring-shaped structures **111** can occur with a single continuous depression of trigger **202**. In all of these embodiments, proximal control device **200** can instead be configured such that repeated pulls of trigger **202** are required to deploy all of ring-shaped structures **111** of implant **102**.

During deployment, e.g., after time T**0** up until completed deployment of the proximal-most ring-shaped structure **112**, if the physician wishes to recapture implant **102**, then depression of trigger **202** can be stopped. Trigger **202** can be spring-loaded or otherwise biased to return to the outermost position. See FIG. **6**B.

If the physician is satisfied with deployment, then at **1014** distal engagement portion **114** and proximal engagement portion **115** of implant **102** can be released from distal control member **140** and grasper **136**, respectively. By way of example, in proximal control device **200** the physician can pull tab **910** to permit trigger **202** to be depressed the rest of the way, which in turn can deploy proximal engagement portion **115** of implant **102**, either by distal advancement of grasper **136**, proximal retraction of shafts **120** and **130**, or both. A tab can be coupled with control wire **146** and the pulling of the tab can pull wire **146** and remove retainer **142** from distal engagement portion **114**.

Anchor **152** can then be recaptured (e.g., deflation of the balloon or retraction of the wire-form members) and withdrawn into anchor delivery member **150** if desired. Anchor delivery member **150**, distal control member **140**, and inner shaft **130** can be retracted into outer shaft **120** and then withdrawn from the urethra.

A more detailed description of the process by which the components in the control device accomplish the above steps is provided in International Application No. PCT/US19/32637, filed May 16, 2019, which was previously incorporated by reference in its entirety for all purposes.

Example Embodiments of User Assembly of Proximal Control Device

Referring back to FIG. **5**A, proximal control device **200** can include a movable (e.g., retractable and/or advanceable) handle portion **1102** that can move with respect to the more proximally located handle portion **1103**. FIG. **5**A depicts movable handle portion **1102** in a distally advanced position prior to deployment of implant **102** and FIG. **5**B depicts portion **1102** in a proximally retracted position after deployment of implant **102**. Movable portion **1102** can be secured to and moved with outer shaft **120**, and can also be moved independently of inner shaft **130**, distal control member **140**, and anchor delivery member **150** (not shown).

A more detailed description of embodiments of the proximal control device is provided in International Application No. PCT/US19/32637, filed May 16, 2019, which was previously incorporated by reference in its entirety for all purposes. Additional details may be found in U.S. Publ. No. 2021/0145619, which is hereby incorporated by reference in its entirety for all purposes.

Example Embodiments of Implant Placement

All the embodiments of system **100** described herein can be used to deliver implant **102** to various locations in proximity to the prostate gland, or other locations within the human anatomy. FIG. **7** is a cross-section of the male anatomy that provides context for use in describing various examples of implantation locations within the prostatic urethra. Here, prostate gland **1302** is centrally located with bladder wall **1304** and bladder **1305** located superiorly. The prostatic urethra **1306** extends inferiorly from bladder **1305** past ejaculatory duct **1307** and through prostate gland **1302**. The prostatic urethra **1306** becomes the membranous urethra **1308** at the general location of the external urethral sphincter **1309** and continues on to exit the body. The rectum is indicated by **1310**.

FIG. **8**A is a cross-section rotated from the viewpoint of FIG. **7** such that the posterior direction extends into the page in the anterior direction extends out of the page. Here an example embodiment of implant **102** is shown positioned within prostatic urethra **1306**. Implant **102** is generally positioned centrally within prostatic urethra **1306** as viewed from this perspective, in other words, generally an equal distance from the superior and inferior edges of prostate gland **1302**. Placement of implant **102** is generally at the discretion of the medical professional and can be offset either superiorly or inferiorly from the positions shown here, however a position within the prostatic urethra **1306** is generally preferred.

FIG. **8**B depicts the area of prostate gland **1302** from generally the same perspective as that of FIG. **7**, but with more detail. Here, prostate gland **1302** is in an enlarged state with a median lobe **1402** that protrudes into prostatic urethra **1306**. FIG. **8**C is a cross-section taken along line **8**C-**8**C of FIG. **8**B and shows the slit-like nature of prostatic urethra **1306** in this enlarged prostate gland **1302** where the width of urethra **1306** widens as it progresses from the anterior to the posterior side.

FIG. **8**D depicts an example embodiment of a posteriorly placed implant **102** within the example anatomy described with respect to FIG. **8**B and FIG. **8**E is a cross-section taken along line **8**E-**8**E of FIG. **8**D. As can be seen here, implant **102** is placed generally along the posterior most surface of the prostatic urethra **1306**. Implant **102** is sized to have a maximum diameter that is less than the width of prostatic urethra **1306** at its maximum central width (e.g., less than 50% of the width, less than 65% of the width, less than 80% of the width, etc.) such that implant **102** can be described as residing substantially on the posterior side of prostatic urethra **1306**, and not in contact with the anterior most side of urethra **1306**. The implications of this placement are shown in FIG. **8**E where the opening through prostate gland **1302** that is created by implant **102** is positioned primarily on the posterior side of prostate gland **1302** and urethra **1306**.

FIG. **8**F depicts an example embodiment of an anteriorly placed implant **102** within the example anatomy described with respect to FIG. **8**B and FIG. **8**G is a cross-section taken along line **8**G-**8**G of FIG. **8**E. As can be seen here, implant **102** is placed generally along the anterior most surface of prostatic urethra **1306**. Implant **102** can be sized to have a maximum diameter that is less than the width of prostatic urethra **1306** at its maximum central width (e.g., less than 50% of the width, less than 65% of the width, less than 80% of the width, etc.) such that implant **102** can be described as residing substantially on the anterior side of prostatic urethra **1306**, and not in contact with the posterior most side of urethra **1306**. The implications of this placement are shown in FIG. **8**G where the opening through prostate gland **1302** that is created by implant **102** is positioned primarily on the anterior side of prostate gland **1302** and urethra **1306**. With both the posterior placement and the anterior placement, implant **102** can still be placed generally centrally with respect to prostate gland **1302** as shown in FIG. **8**A. Deployment of implant **102** in a posterior or anterior position is generally at the discretion of the medical professional. Other variations of placement can also be used including placements that are centrally located between the posterior most side and interior most side of urethra **1306**, as well as variations in sizing such that implant **102** has a relatively larger or smaller diameter with respect to prostate **1302** than shown here.

Removal Methods

In one embodiment, the implant **102** can be removed by grasping an atraumatic end of the device and deforming the device from its expanded spiral shape to a roughly linear shape, such that it can be withdrawn through the lumen of a catheter. A retrieval device **300** configured to be inserted through the lumen **121** of outer tubular member **120** of delivery device **103** can be used for both acute and chronic retrieval of the implant **102**. As depicted in FIGS. **9**A-**9**C, delivery device **103** can be used to remove implant **102**. An elongate flexible tubular member **330** can be delivered through first inner lumen **121** of outer tubular member **120**. The elongate flexible tubular member **330** can be made of a polymer, e.g., PEEK. An actuating shaft **334** can be delivered through a lumen of the elongate flexible tubular member **330**. The actuating shaft **334** may have forcep cups or opposed jaws **338***a, b* at a distal end and an actuating handle **336** configured to manipulate the opening and closing of the opposed jaws **338***a, b* at a proximal end. The actuating shaft **334** may also be configured to move axially relative to the elongate flexible tubular member **330** when the device handle **336** is articulated. The distal end of the actuating shaft **334** can be extended past the distal end of the elongate flexible tubular member **330**, which can be extended past the distal end of the outer tubular member **120**, to grasp either of atraumatic ends **116**, **117**, which are, e.g., rounded, spherical, cylindrical, frustoconical, or ballized of the implant **102**. The opposed jaws **338***a, b* are configured to hold one of atraumatic ends **116**, **117** within a cavity defined by the opposed jaws **338***a, b* in a closed configuration. As seen in FIGS. **10**A-**10**F, the opposed jaws **338***a, b* may each further include a curved edge **342***a, b* in a distal region of the opposed jaws **338***a, b*, such that an opening **340** in the distal end is formed when the opposed jaws **338***a, b* are in the closed configuration, where the opening **340** communicates with the cavity defined by the opposed jaws **338***a, b* in the closed configuration. As seen in FIGS. **11**A-**11**B, once either of atraumatic ends **116**, **117** are grasped, proximal withdrawal of the articulating shaft **334** into the lumen of elongate flexible tubular member **330** will result in linearizing the implant **102** (i.e., pulling the implant into a roughly linear form) such that the implant can be withdrawn through the lumen **121** of outer tubular member **120**.

In another embodiment, the implant can be removed by grasping anywhere along the length of the implant, rather than only grasping an atraumatic end. Such a procedure could be performed in an in-office outpatient procedure. A single-use retrieval device **400** configured to be inserted through the lumen **121** of outer tubular member **120** of delivery device **103** can be used for both acute and chronic retrieval of the implant **102**. As seen in FIG. **12**, retrieval device **400** includes an articulating handle **410**, semi-flexible sheath **414**, actuating shaft **418** with grasping hook **420** at the distal end. Semi-flexible sheath **414** may be a stainless steel tube with a laser cut pattern in a distal region **416** for increased flexibility. The distal region **416** may be from about 3 to about 8 inches, alternatively from about 3 to about 7 inches, alternatively from about 4 to about 7 inches, alternatively from about 3 to about 6 inches from a distal end of the semi-flexible sheath **414**. The laser cut pattern can allow for maneuverability when used through the working length of the delivery device **103**. Within the semi-flexible sheath **414** is an actuating shaft **418** that is configured to move axially relative to the sheath **410** when the device handle **410** is articulated. The actuating shaft **418** may also be semi-flexible such that it can flex and maneuver when used through the working channel of the delivery device **103**. Grasping hook **420** is located at the distal end of the actuating shaft **418**. The atraumatic grasping hook **420** extends beyond the distal end of sheath **414** and can be visible via an imaging device when the retrieval device handle is in the "open" position. The grasping hook **420** can be withdrawn and recessed into sheath **414** when the articulating handle **410** is in the "closed" position.

As seen in FIGS. **15A-15C**, the hook **420** may be shaped to enable visualization of the implant. As seen in FIG. **15C**, a tip **428** of the hook may be narrower than a back portion **430** of the hook. The hook **420** may have a cross-sectional shape in which a back-side portion **430** of the cross-sectional shape is wider than a front-side portion **428**. A width of the tip **428** of the hook may be between about 0.015" to about 0.050", alternatively between about 0.010" to about 0.040", alternatively between about 0.01" to about 0.030". The width of the front side portion **428** of the hook may be about ⅓ to about ⅔, alternatively about ¼ to about ¾, alternatively about ½ narrower than the largest width of the back-side portion **430**.

Sheath **414** of retrieval device **400** is intended to be inserted into the first inner lumen **121** of outer tubular member **120** and maneuvered to the implant **102** to be removed. The grasping hook **420** may grasp any portion of the implant **102**. For instance, the grasping hook **420** may grasp the implant anywhere along the elongate wire of the implant **102**. The implant **102** has been "grasped" when any portion of it is inside the slot of the grasping hook **420** (see FIGS. **13A-13D**). Once the implant is engaged by the grasping hook **420**, the user may articulate the device handle **410**. As seen in FIG. **14**, articulation of the device handle **410** from an open to a closed position applies a 3-point bend to the grasped portion of the implant. The 3-point bend may be caused by force applied to the implant **102** by the outside wall **422** of the sheath **414** (see arrows in direction B) as the grasping hook **420** is withdrawn into the sheath **414** (see arrow in direction A). As force is applied to the device handle and the grasping hook is retracted, the implant effectively crimps due to the force of the 3-point bend. The grasping hook **420** and implant **102** are also partially withdrawn into the semi-flexible sheath. This action can induce high strain, which can permanently deform the implant. Once the handle has been fully articulated to a closed position and the implant is crimped and partially withdrawn the sheath **414**, the implant **102** can be pulled from the lumen **121** and removed from the patient. The mechanism of removal relies on a "double lineation" of the implant; the wire on each side of the crimped hook straightens to substantially linear shapes as the retrieval device is pulled backward through the scope working channel. The sheath **414** may be a hypotube, e.g., a lasercut hypotube. In other embodiments, the sheath **414** may include a rigid member disposed over a distal region of the sheath **414**, where the rigid member has compressive support and is sufficiently rigid to apply two counter forces to the implant **102** to form the 3-point bend. The rigid member may be tubular or non-tubular in shape.

The embodiments described herein are restated and expanded upon in the following paragraphs without explicit reference to the figures.

In many embodiments, a method of removing an implant from the urethra of a patient is described, the method including: advancing a portion of a removal device within a urethra of a patient to a position near the implant, wherein the removal device comprises an outer tubular member, an inner elongate tubular member within a lumen of the outer tubular member, and an elongate actuating member within a lumen of the inner elongate tubular member, the elongate actuating member having a proximal end coupled to a handle and a distal end comprising a hook; advancing the distal end of the elongate actuating member distal of a distal end of the outer tubular member and distal of a distal end of the inner elongate tubular member; grasping a portion of the implant with the hook; proximally withdrawing the hook and at least a portion of the implant into the lumen of the inner elongate tubular member such that a bend forms in the implant; and withdrawing the implant through the lumen of the outer tubular member, wherein the implant is in an axially elongate shape while in the lumen of the inner elongate tubular member.

In some embodiments, the implant has a first end, a second end, and a middle portion, and wherein the hook grasps the implant in the middle portion.

In some embodiments, the implant comprises an elongate wire and at least one enlarged end, and wherein the hook grasps the elongate wire.

In some embodiments, the inner elongate tubular member comprises a flexible distal region. In some embodiments, the flexible distal region comprises a laser cut tube.

In some embodiments, the inner elongate tubular member is a hypotube.

In some embodiments, the removal device further includes a rigid member disposed over a distal region of the inner elongate tubular member.

In some embodiments, a three-point bend forms in the implant.

In some embodiments, a bend forms in the implant as a result of a force applied to the implant by a wall of the inner elongate tubular member and the hook.

In some embodiments, the axially elongate shape includes a double lineation of the implant.

In some embodiments, the method further includes the step of viewing the hook with an imaging device after the hook is advanced distally beyond the distal end of the outer tubular member.

In some embodiments, the hook and the at least a portion of the implant are proximally withdrawn by actuating the handle coupled to the proximal end of the elongate actuating member.

In some embodiments, the implant has an expanded helical shape.

In some embodiments, a tip of the hook has a width between about 0.015" to about 0.050".

In some embodiments, a tip portion of the hook is narrower than a back portion of the hook.

In many embodiments, a system for retrieving an implant is described, the system includes a retrieval device including: an outer tubular member having a distal end and a lumen; an inner elongate tubular member within the lumen of the outer tubular member; an elongate actuating member within a lumen of the inner elongate tubular member, the elongate actuating member having a proximal end coupled to a handle and a distal end comprising a hook; and a proximal control device coupled with the inner tubular member and releasably coupled with the outer tubular member through a coupling mechanism, where the proximal control device is configured to longitudinally move the inner elongate tubular member, the outer tubular member, and the elongate actuating member, and also configured to move the elongate actuating member within the lumen of the inner elongate tubular member.

In some embodiments, the implant has a first end, a second end, and a middle portion, and wherein the hook is configured to grasp the implant in the middle portion.

In some embodiments, the inner elongate tubular member comprises a flexible distal region. In some embodiments, the flexible distal region comprises a laser cut tube.

In some embodiments, the inner elongate tubular member is a hypotube.

In some embodiments, the system further includes a rigid member disposed over a distal region of the inner elongate tubular member.

In some embodiments, the outer tubular member further comprises an imaging device located in a distal end region of the outer tubular member.

In some embodiments, the proximal control device is configured to longitudinally move the inner elongate tubular member, the outer tubular member, and the elongate actuating member concurrently.

In some embodiments, the implant has an expanded helical shape.

In many embodiments, a method of removing an implant from the urethra of a patient is described, the method including: advancing a removal device within a urethra of a patient to a position near the implant, wherein the removal device comprises an outer tubular member, an inner elongate tubular member within a lumen of the outer tubular member, an elongate actuating member within a lumen of the inner elongate tubular member, the elongate actuating member comprising a proximal end and a distal end, wherein the distal end comprises opposed first and second jaws configured to open and close; advancing the distal end of the elongate actuating member distal of a distal end of the inner elongate tubular member; grasping a portion of the implant with the opposed first and second jaws; and proximally withdrawing the elongate actuating member in the lumen of the inner elongate tubular member, wherein at least a portion of the implant assumes an axially elongated shape within the lumen of the inner elongate tubular member.

In some embodiments, the opposed first and second jaws grasp an enlarged end of the implant. In some embodiments, the shape of the enlarged end is can be a ball, a cylinder, or a cone. In some embodiments, the opposed first and second jaws are in a closed configuration, the opposed first and second jaws have an opening at a distal end of the closed configuration. In some embodiments, the implant includes an elongate wire and at least one enlarged end, and where the at least one enlarged end is grasped by the opposed first and second jaws, and where the elongate wire extends through the opening at the distal end of the closed configuration.

In some embodiments, the implant is in an axially elongate shape while in the lumen of the outer tubular member. In some embodiments, the axially elongated shape is substantially linear.

In some embodiments, the method further includes the step of viewing the distal end of the elongate actuating member with an imaging device after the distal end of the elongate actuating member is advanced distally beyond the distal end of the outer tubular member.

In some embodiments, the implant has an expanded helical shape.

In some embodiments, a tip of the hook has a width between about 0.015" to about 0.050".

In some embodiments, a tip portion of the hook is narrower than a back portion of the hook.

In many embodiments, a system for retrieving an implant is described, the system including a retrieval device including: an outer tubular member having a distal end and a lumen; an inner elongate tubular member within the lumen of the outer tubular member; an elongate actuating member within a lumen of the inner elongate tubular member, the elongate actuating member comprising a proximal end and a distal end, wherein the distal end comprises first and second jaws configured to open and close; and a proximal control device coupled with the elongate actuating member and the inner elongate tubular member, and releasably coupled with the outer tubular member through a coupling mechanism, where the proximal control device is configured to longitudinally move the elongate actuating member, inner elongate tubular member, and the outer tubular member, and also configured to move the elongate actuating member longitudinally within the lumen of the inner tubular member.

In some embodiments, the implant has a first end, a second end, and a middle portion, and wherein at least one of the first and second ends is an enlarged atraumatic end. In some embodiments, the opposed first and second jaws are configured to grasp the enlarged atraumatic end of the implant.

In some embodiments, the opposed first and second jaws, when in a closed configuration, have an opening at a distal end of the closed configuration. In some embodiments, the implant comprises an elongate wire and at least one enlarged end, and wherein the opposed first and second jaws are configured to grasp the at least one enlarged end, and wherein the opening at the distal end of the closed configuration is configured for the elongate wire to extend therethrough.

In some embodiments, the outer tubular member further comprises an imaging device located in a distal end region of the outer tubular member.

In some embodiments, the proximal control device is configured to longitudinally move the elongate member, inner elongate member, and the outer tubular member concurrently.

In some embodiments, the implant has an expanded helical shape.

In many embodiments, a method of removing an implant from the urethra of a patient is described, the method including: advancing a removal device within a urethra of a patient to a position near the implant, wherein the removal device comprises an outer tubular member, an inner elongate tubular member within a lumen of the outer tubular member, an elongate actuating member within a lumen of the inner elongate tubular member, the elongate actuating member comprising a proximal end and a distal end, wherein the distal end comprises a grasper; advancing the distal end of the elongate actuating member distal of a distal end of the inner elongate tubular member; grabbing a portion of the implant with the grasper; and proximally withdrawing the elongate actuating member in the lumen of the inner elongate tubular member, wherein at least a portion of the implant assumes an axially elongated shape within the lumen of the inner elongate tubular member.

In some embodiments, the grasper comprises opposed first and second jaws. In some embodiments, the opposed first and second jaws grab an enlarged end of the implant. In some embodiments, the opposed first and second jaws are in a closed configuration, the opposed first and second jaws have an opening at a distal end of the closed configuration. In some embodiments, the implant comprises an elongate wire and at least one enlarged end, and wherein the at least one enlarged end is grabbed by the opposed first and second jaws, and wherein the elongate wire extends through the opening at the distal end of the closed configuration. In some embodiments, the axially elongated shape is substantially linear.

In some embodiments, the grasper comprises a hook. In some embodiments, the implant has a first end, a second end, and a middle portion, and wherein the hook grasps the implant in the middle portion. In some embodiments, the implant comprises an elongate wire and at least one enlarged end, and wherein the hook grasps the elongate wire. In some embodiments, the inner elongate tubular member comprises a flexible distal region. In some embodiments, the flexible distal region comprises a laser cut tube. In some embodiments, the inner elongate tubular member is a hypotube. In some embodiments, the removal device further includes a rigid member disposed over a distal region of the inner elongate tubular member. In some embodiments, when the hook and the portion of the implant are proximally withdrawn in the lumen of the inner elongate tubular member, a bend forms in the implant. In some embodiments, the bend is a three-point bend. In some embodiments, the bend forms in the implant as a result of a plurality of forces applied to the implant by a wall of the inner elongate tubular member and the hook. In some embodiments, the axially elongate shape includes a double lineation of the implant. In some embodiments, the method further includes the step of viewing the hook with an imaging device after the hook is advanced distally beyond the distal end of the outer tubular member.

In some embodiments, the implant has an expanded helical shape.

In some embodiments, a tip of the hook has a width between about 0.015" to about 0.050".

In some embodiments, a tip portion of the hook is narrower than a back portion of the hook.

In many embodiments, a system for retrieving an implant is described, the system including a retrieval device including: an outer tubular member having a distal end and a lumen; an inner elongate tubular member within the lumen of the outer tubular member; an elongate actuating member within a lumen of the inner elongate tubular member, the elongate actuating member comprising a proximal end and a distal end, wherein the distal end comprises a grasper; and a proximal control device coupled with the elongate actuating member and the inner elongate actuating member, and releasably coupled with the outer tubular member through a coupling mechanism, wherein the proximal control device is configured to longitudinally move the elongate actuating member, inner elongate tubular member, and the outer tubular member, and also configured to move the elongate actuating member longitudinally within the lumen of the inner tubular member.

In some embodiments, the grasper comprises opposed first and second jaws. In some embodiments, the implant has a first end, a second end, and a middle portion, and wherein at least one of the first and second ends is an enlarged atraumatic end. In some embodiments, the opposed first and second jaws are configured to grasp the enlarged atraumatic end of the implant. In some embodiments, the opposed first and second jaws, when in a closed configuration, have an opening at a distal end of the closed configuration. In some embodiments, the implant comprises an elongate wire and at least one enlarged end, and wherein the opposed first and second jaws are configured to grasp the at least one enlarged end, and wherein the opening at the distal end of the closed configuration is configured for the elongate wire to extend therethrough.

In some embodiments, the outer tubular member further comprises an imaging device located in a distal end region of the outer tubular member.

In some embodiments, the proximal control device is configured to longitudinally move the elongate member, inner elongate member, and the outer tubular member concurrently.

In some embodiments, the grasper comprises a hook. In some embodiments, the implant has a first end, a second end, and a middle portion, and wherein the hook is configured to grasp the implant in the middle portion. In some embodiments, the inner elongate tubular member comprises a flexible distal region. In some embodiments, the inner elongate tubular member is a hypotube. In some embodiments, the system further includes a rigid member disposed over a distal region of the inner elongate tubular member. In some embodiments, the flexible distal region comprises a laser cut tube. In some embodiments, the implant has an expanded helical shape. In some embodiments, a tip of the hook has a width between about 0.015" to about 0.050". In some embodiments, a tip portion of the hook is narrower than a back portion of the hook.

Systems, devices, and methods are provided for retrieval of an implant from the prostatic urethra. Embodiments of retrieval systems can include a device for insertion into the patient and a proximal control device for use in grasping a portion of the implant and withdrawing the implant into a lumen of the retrieval system.

Aspects of the invention are set out in the independent claims and preferred features are set out in the dependent claims. Preferred features of each aspect may be provided in combination with each other within particular embodiments and may also be provided in combination with other aspects.

Clauses

Exemplary embodiments are set out in the following numbered clauses.

Clause 1. A method of removing an implant from the urethra of a patient, the method comprising: advancing a portion of a removal device within a urethra of a patient to a position near the implant, wherein the removal device comprises an outer tubular member, an inner elongate tubular member within a lumen of the outer tubular member, and an elongate actuating member within a lumen of the inner elongate tubular member, the elongate actuating member having a proximal end coupled to a handle and a distal end comprising a hook; advancing the distal end of the elongate actuating member distal of a distal end of the outer tubular member and distal of a distal end of the inner elongate tubular member; grasping a portion of the implant with the hook; proximally withdrawing the hook and at least a portion of the implant into the lumen of the inner elongate tubular member such that a bend forms in the implant; and withdrawing the implant through the lumen of the outer tubular member, wherein the implant is in an axially elongate shape while in the lumen of the inner elongate tubular member.

Clause 2. The method of clause 1, wherein the implant has a first end, a second end, and a middle portion, and wherein the hook grasps the implant in the middle portion.

Clause 3. The method of clause 1, wherein the implant comprises an elongate wire and at least one enlarged end, and wherein the hook grasps the elongate wire.

Clause 4. The method of clause 1, wherein the inner elongate tubular member comprises a flexible distal region.

Clause 5. The method of clause 4, wherein the flexible distal region comprises a laser cut tube.

Clause 6. The method of clause 1, wherein the inner elongate tubular member is a hypotube.

Clause 7. The method of clause 1, wherein the removal device further comprises a rigid member disposed over a distal region of the inner elongate tubular member.

Clause 8. The method of clause 1, wherein a three-point bend forms in the implant.

Clause 9. The method of clause 1, wherein a bend forms in the implant as a result of a force applied to the implant by a wall of the inner elongate tubular member and the hook.

Clause 10. The method of clause 1, wherein the axially elongate shape includes a double lineation of the implant.

Clause 11. The method of clause 1, further comprising the step of viewing the hook with an imaging device after the hook is advanced distally beyond the distal end of the outer tubular member.

Clause 12. The method of clause 1, wherein the hook and the at least a portion of the implant are proximally withdrawn by actuating the handle coupled to the proximal end of the elongate actuating member.

Clause 13. The method of clause 1, wherein the implant has an expanded helical shape.

Clause 14. The method of clause 1, wherein a tip of the hook has a width between about 0.015" to about 0.050".

Clause 15. The method of clause 1, wherein a tip portion of the hook is narrower than a back portion of the hook.

Clause 16. A system for retrieving an implant, the system comprising a retrieval device comprising: an outer tubular member having a distal end and a lumen; an inner elongate tubular member within the lumen of the outer tubular member; an elongate actuating member within a lumen of the inner elongate tubular member, the elongate actuating member having a proximal end coupled to a handle and a distal end comprising a hook; and a proximal control device coupled with the inner tubular member and releasably coupled with the outer tubular member through a coupling mechanism, wherein the proximal control device is configured to longitudinally move the inner elongate tubular member, the outer tubular member, and the elongate actuating member, and also configured to move the elongate actuating member within the lumen of the inner elongate tubular member.

Clause 17. The system of clause 16, wherein the implant has a first end, a second end, and a middle portion, and wherein the hook is configured to grasp the implant in the middle portion.

Clause 18. The system of clause 16, wherein the inner elongate tubular member comprises a flexible distal region.

Clause 19. The system of clause 16, wherein the inner elongate tubular member is a hypotube.

Clause 20. The system of clause 16, wherein the system further comprises a rigid member disposed over a distal region of the inner elongate tubular member.

Clause 21. The system of clause 18, wherein the flexible distal region comprises a laser cut tube.

Clause 22. The system of clause 16, wherein the outer tubular member further comprises an imaging device located in a distal end region of the outer tubular member.

Clause 23. The system of clause 16, wherein the proximal control device is configured to longitudinally move the inner elongate tubular member, the outer tubular member, and the elongate actuating member concurrently.

Clause 24. The system of clause 16, wherein the implant has an expanded helical shape.

Clause 25. The system of clause 16, wherein a tip of the hook has a width between about 0.015" to about 0.050".

Clause 26. The system of clause 16, wherein a tip portion of the hook is narrower than a back portion of the hook.

Clause 27. A method of removing an implant from the urethra of a patient, the method comprising: advancing a removal device within a urethra of a patient to a position near the implant, wherein the removal device comprises an outer tubular member, an inner elongate tubular member within a lumen of the outer tubular member, an elongate actuating member within a lumen of the inner elongate tubular member, the elongate actuating member comprising a proximal end and a distal end, wherein the distal end comprises opposed first and second jaws configured to open and close; advancing the distal end of the elongate actuating member distal of a distal end of the inner elongate tubular member; grasping a portion of the implant with the opposed first and second jaws; and proximally withdrawing the elongate actuating member in the lumen of the inner elongate tubular member, wherein at least a portion of the implant assumes an axially elongated shape within the lumen of the inner elongate tubular member.

Clause 28. The method of clause 27, wherein the opposed first and second jaws grasp an enlarged end of the implant.

Clause 29. The method of clause 28, wherein the enlarged end is in a shape selected from the group consisting of a ball, a cylinder, and a cone.

Clause 30. The method of clause 28, wherein when the opposed first and second jaws are in a closed configuration, the opposed first and second jaws have an opening at a distal end of the closed configuration.

Clause 31. The method of clause 30, wherein the implant comprises an elongate wire and at least one enlarged end, and wherein the at least one enlarged end is grasped by the opposed first and second jaws, and wherein the elongate wire extends through the opening at the distal end of the closed configuration.

Clause 32. The method of clause 27, wherein the implant is in an axially elongate shape while in the lumen of the outer tubular member.

Clause 33. The method of clause 32, wherein the axially elongated shape is substantially linear.

Clause 34. The method of clause 27, further comprising the step of viewing the distal end of the elongate actuating member with an imaging device after the distal end of the elongate actuating member is advanced distally beyond the distal end of the outer tubular member.

Clause 35. The method of clause 27, wherein the implant has an expanded helical shape.

Clause 36. A system for retrieving an implant, the system comprising a retrieval device comprising: an outer tubular member having a distal end and a lumen; an inner elongate tubular member within the lumen of the outer tubular member; an elongate actuating member within a lumen of the inner elongate tubular member, the elongate actuating member comprising a proximal end and a distal end, wherein the distal end comprises first and second jaws configured to open and close; and a proximal control device coupled with the elongate actuating member and the inner elongate tubular member, and releasably coupled with the outer tubular member through a coupling mechanism, wherein the proximal control device is configured to longitudinally move the elongate actuating member, inner elongate tubular member, and the outer tubular member, and also configured to move the elongate actuating member longitudinally within the lumen of the inner tubular member.

Clause 37. The system of clause 36, wherein the implant has a first end, a second end, and a middle portion, and wherein at least one of the first and second ends is an enlarged atraumatic end.

Clause 38. The system of clause 37, wherein the opposed first and second jaws are configured to grasp the enlarged atraumatic end of the implant.

Clause 39. The system of clause 36, wherein the opposed first and second jaws, when in a closed configuration, have an opening at a distal end of the closed configuration.

Clause 40. The system of clause 39, wherein the implant comprises an elongate wire and at least one enlarged end, and wherein the opposed first and second jaws are configured to grasp the at least one enlarged end, and wherein the opening at the distal end of the closed configuration is configured for the elongate wire to extend therethrough.

Clause 41. The system of clause 36, wherein the outer tubular member further comprises an imaging device located in a distal end region of the outer tubular member.

Clause 42. The system of clause 36, wherein the proximal control device is configured to longitudinally move the elongate member, inner elongate member, and the outer tubular member concurrently.

Clause 43. The system of clause 36, wherein the implant has an expanded helical shape.

Clause 44. A method of removing an implant from the urethra of a patient, the method comprising: advancing a removal device within a urethra of a patient to a position near the implant, wherein the removal device comprises an outer tubular member, an inner elongate tubular member within a lumen of the outer tubular member, an elongate actuating member within a lumen of the inner elongate tubular member, the elongate actuating member comprising a proximal end and a distal end, wherein the distal end comprises a grasper; advancing the distal end of the elongate actuating member distal of a distal end of the inner elongate tubular member; grabbing a portion of the implant with the grasper; and proximally withdrawing the elongate actuating member in the lumen of the inner elongate tubular member, wherein at least a portion of the implant assumes an axially elongated shape within the lumen of the inner elongate tubular member.

Clause 45. The method of clause 44, wherein the grasper comprises opposed first and second jaws.

Clause 46. The method of clause 45, wherein the opposed first and second jaws grab an enlarged end of the implant.

Clause 47. The method of clause 45, wherein when the opposed first and second jaws are in a closed configuration, the opposed first and second jaws have an opening at a distal end of the closed configuration.

Clause 48. The method of clause 47, wherein the implant comprises an elongate wire and at least one enlarged end, and wherein the at least one enlarged end is grabbed by the opposed first and second jaws, and wherein the elongate wire extends through the opening at the distal end of the closed configuration.

Clause 49. The method of clause 45, wherein the axially elongated shape is substantially linear.

Clause 50. The method of clause 44, wherein the grasper comprises a hook.

Clause 51. The method of clause 50, wherein the implant has a first end, a second end, and a middle portion, and wherein the hook grasps the implant in the middle portion.

Clause 52. The method of clause 50, wherein the implant comprises an elongate wire and at least one enlarged end, and wherein the hook grasps the elongate wire.

Clause 53. The method of clause 50, wherein a tip of the hook has a width between about 0.015" to about 0.050".

Clause 54. The method of clause 50, wherein a tip portion of the hook is narrower than a back portion of the hook.

Clause 55. The method of clause 50, wherein the inner elongate tubular member comprises a flexible distal region.

Clause 56. The method of clause 55, wherein the flexible distal region comprises a laser cut tube.

Clause 57. The method of clause 50, wherein the inner elongate tubular member is a hypotube.

Clause 58. The method of clause 50, wherein the removal device further comprises a rigid member disposed over a distal region of the inner elongate tubular member.

Clause 59. The method of clause 50, wherein when the hook and the portion of the implant are proximally withdrawn in the lumen of the inner elongate tubular member, a bend forms in the implant.

Clause 60. The method of clause 59, wherein the bend is a three-point bend.

Clause 61. The method of clause 59, wherein the bend forms in the implant as a result of a plurality of forces applied to the implant by a wall of the inner elongate tubular member and the hook.

Clause 62. The method of clause 50, wherein the axially elongate shape includes a double lineation of the implant.

Clause 63. The method of clause 50, further comprising the step of viewing the hook with an imaging device after the hook is advanced distally beyond the distal end of the outer tubular member.

Clause 64. The method of clause 44, wherein the implant has an expanded helical shape.

Clause 65. A system for retrieving an implant, the system comprising a retrieval device comprising: an outer tubular member having a distal end and a lumen; an inner elongate tubular member within the lumen of the outer tubular member; an elongate actuating member within a lumen of the inner elongate tubular member, the elongate actuating member comprising a proximal end and a distal end, wherein the distal end comprises a grasper; and a proximal control device coupled with the elongate actuating member and the inner elongate actuating member, and releasably coupled with the outer tubular member through a coupling mechanism, wherein the proximal control device is configured to longitudinally move the elongate actuating member, inner elongate tubular member, and the outer tubular member, and also configured to move the elongate actuating member longitudinally within the lumen of the inner tubular member.

Clause 66. The system of clause 65, wherein the grasper comprises opposed first and second jaws.

Clause 67. The system of clause 66, wherein the implant has a first end, a second end, and a middle portion, and wherein at least one of the first and second ends is an enlarged atraumatic end.

Clause 68. The system of clause 67, wherein the opposed first and second jaws are configured to grasp the enlarged atraumatic end of the implant.

Clause 69. The system of clause 66, wherein the opposed first and second jaws, when in a closed configuration, have an opening at a distal end of the closed configuration.

Clause 70. The system of clause 69, wherein the implant comprises an elongate wire and at least one enlarged end, and wherein the opposed first and second jaws are configured to grasp the at least one enlarged end, and wherein the opening at the distal end of the closed configuration is configured for the elongate wire to extend therethrough.

Clause 71. The system of clause 65, wherein the outer tubular member further comprises an imaging device located in a distal end region of the outer tubular member.

Clause 72. The system of clause 65, wherein the proximal control device is configured to longitudinally move the elongate member, inner elongate member, and the outer tubular member concurrently.

Clause 73. The system of clause 65, wherein the grasper comprises a hook.

Clause 74. The system of clause 73, wherein the implant has a first end, a second end, and a middle portion, and wherein the hook is configured to grasp the implant in the middle portion.

Clause 75. The system of clause 73, wherein a tip of the hook has a width between about 0.01 Clause 5" to about 0.050".

Clause 76. The system of clause 73, wherein a tip portion of the hook is narrower than a back portion of the hook.

Clause 77. The system of clause 73, wherein the inner elongate tubular member comprises a flexible distal region.

Clause 78. The system of clause 77, wherein the flexible distal region comprises a laser cut tube.

Clause 79. The system of clause 73, wherein the inner elongate tubular member is a hypotube.

Clause 80. The system of clause 73, wherein the system further comprises a rigid member disposed over a distal region of the inner elongate tubular member.

Clause 81. The system of clause 65, wherein the implant has an expanded helical shape.

All features, elements, components, functions, and steps described with respect to any embodiment provided herein are intended to be freely combinable and substitutable with those from any other embodiment. If a certain feature, element, component, function, or step is described with respect to only one embodiment, then it should be understood that that feature, element, component, function, or step can be used with every other embodiment described herein unless explicitly stated otherwise. This paragraph therefore serves as antecedent basis and written support for the introduction of claims, at any time, that combine features, elements, components, functions, and steps from different embodiments, or that substitute features, elements, components, functions, and steps from one embodiment with those of another, even if the following description does not explicitly state, in a particular instance, that such combinations or substitutions are possible. It is explicitly acknowledged that express recitation of every possible combination and substitution is overly burdensome, especially given that the permissibility of each and every such combination and substitution will be readily recognized by those of ordinary skill in the art.

As used herein and in the appended claims, the singular forms "a," "an," and "the" include plural referents unless the context clearly dictates otherwise.

While the embodiments are susceptible to various modifications and alternative forms, specific examples thereof have been shown in the drawings and are herein described in detail. It should be understood, however, that these embodiments are not to be limited to the particular form disclosed, but to the contrary, these embodiments are to cover all modifications, equivalents, and alternatives falling within the spirit of the disclosure. Furthermore, any features, functions, steps, or elements of the embodiments may be recited in or added to the claims, as well as negative limitations that define the inventive scope of the claims by features, functions, steps, or elements that are not within that scope.

What is claimed is:

1. A method of removing an implant from the urethra of a patient, the method comprising:

advancing a removal device within a urethra of a patient to a position near the implant, wherein the implant comprises a first portion, a second portion, and a third portion between the first portion and the second portion, wherein the removal device comprises an outer tubular member, an inner elongate tubular member within a lumen of the outer tubular member, an elongate actuating member within a lumen of the inner elongate tubular member, the elongate actuating member comprising a proximal end and a distal end, wherein the distal end comprises a grasper;

advancing the distal end of the elongate actuating member distal of a distal end of the inner elongate tubular member;

grabbing the third portion of the implant with the grasper; and

proximally withdrawing the elongate actuating member in the lumen of the inner elongate tubular member forming a bend in the third portion, wherein the first portion and the second portion of the implant are substantially straightened on either side of the bend within the lumen of the inner elongate tubular member.

2. The method of claim **1**, wherein the grasper comprises a hook.

3. The method of claim **2**, wherein the implant comprises an elongate wire and at least one enlarged end, and wherein the hook grasps the elongate wire.

4. The method of claim **2**, wherein a tip of the hook has a width between about 0.015" to about 0.050".

5. The method of claim **2**, wherein a tip portion of the hook is narrower than a back portion of the hook.

6. The method of claim **2**, wherein the inner elongate tubular member comprises a flexible distal region.

7. The method of claim **6**, wherein the flexible distal region comprises a laser cut tube.

8. The method of claim **2**, wherein the inner elongate tubular member is a hypotube.

9. The method of claim **2**, wherein the removal device further comprises a rigid member disposed over a distal region of the inner elongate tubular member.

10. The method of claim **2**, further comprising the step of viewing the hook with an imaging device after the hook is advanced distally beyond the distal end of the outer tubular member.

11. The method of claim **1**, wherein the bend forms in the implant as a result of a plurality of forces applied to the implant by a wall of the inner elongate tubular member and the hook.

12. The method of claim **1**, wherein the implant has an expanded helical shape.

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